Mandated versus Voluntary Adaptation to Natural Disasters: The Case of US Wildfires

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Despite escalating losses in climate-related disasters, adoption of protective technologies and behaviors is limited by risk misperception, externalities, and insurance market frictions. One response to these market failures is to mandate these investments. We measure the effect of California's wildfire building codes on own and neighboring structure survival using comprehensive data on US homes exposed to wildfires since 2000. Differences across jurisdictions and vintages reveal remarkable resilience effects of building codes. Codes also increase survival of neighboring homes by reducing structure-to-structure spread. We then develop and estimate a model of social benefits of mandatory building standards versus other adaptation policies.

I. Introduction

Take-up of many socially desirable investments and behaviors is thought to be limited because of positive externalities, imperfect information, or other frictions. One response to such barriers is to provide information

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and subsidies to encourage voluntary adoption. An alternative approach is to mandate desired outcomes through minimum standards or bans on alternatives. Mandates ensure wide adoption but may compel some individuals to choose options they would have preferred to avoid even if fully informed and fully accountable. This tension between increased adoption benefits and potentially inefficient adoption exists across a broad range of policies, including energy efficiency standards; regulation of smoking, drinking, and other health behaviors; consumer financial protections; and product and worker safety regulation.[1]

The same trade-off urgently confronts policies intended to guide adaptation to worsening climate-related disasters. Worldwide natural disaster losses averaged US$218 billion per year during 2016–20, a 60% increase in real terms over the preceding 30 years.[2] Efficient investment in adaptation is essential in the face of these escalating risks. Yet efficient take-up of protective technologies and behaviors appears to be hindered by a variety of market frictions: spatial spillovers of mitigation benefits, insurance market imperfections, postdisaster relief programs that soften ex ante incentives, and widespread misperception of risks and benefits.[3] In response, some hazard-prone jurisdictions have encouraged resilience through information campaigns and voluntary incentives.[4] Other

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[1] See, e.g., Jacobsen and Kotchen (2013) and Allcott and Taubinsky (2015) on energy efficiency, Cawley and Ruhm (2011) on health behaviors, Campbell et al. (2011) on financial regulation, and Viscusi (1992) on product and worker safety regulation.

[2] Loss data are from Munich RE and are in 2020 US dollars.

[3] There is growing evidence of these frictions. Benefits of mitigation against some hazards are thought to extend to neighboring properties (Shafran 2008; Lueck and Yoder 2016; Costello, Quérou, and Tomini 2017). Monitoring costs and other insurance market imperfections mean that mitigation behaviors are not accurately reflected in insurance prices (Kunreuther and Michel-Kerjan 2011; California Department of Insurance 2018a; Wagner 2022). Public disaster relief programs may reduce private incentives for property protection (Kousky, Luttmer, and Zeckhauser 2006; Deryugina 2017; Baylis and Boomhower 2023). Hedonic studies imply that homeowners do not accurately perceive disaster risks and thus the value of protective investments (Hallstrom and Smith 2005; Gallagher 2014; McCoy and Walsh 2018; Bakkensen and Barrage 2021).

[4] Examples in the United States include the Ready campaign and https://www.Ready.gov; the Community Rating System under the National Flood Insurance Program; the StormReady, Hurricane Protection Week, and National Tsunami Hazard Mitigation programs; the Firewise USA program; and the Community Wildfire Protection Plan program.

policymakers have instead implemented adaptation mandates, for example, construction standards that require disaster-resistant building materials.[5] In the contest between these approaches, the stakes are high. Well-targeted investments in resilience may prevent vast losses. But such investments are costly, and the net benefits of adaptation mandates depend on how accurately regulators judge the effectiveness of the required actions, the level of the hazard, and individual risk preferences. Moreover, implementing mandatory standards is often politically challenging. Do the additional benefits justify the expenditure of time and political capital? Despite these important questions, there is little empirical evidence about outcomes under a mandated disaster resilience regime compared with a counterfactual of voluntary take-up.

This paper considers the case of wildfire building codes in California, a high-profile adaptation mandate. California suffered nearly US$50 billion in wildfire property losses alone between 2017 and 2021 (Paci, Newman, and Gage 2023). The state also has among the strictest wildfire building codes in the world. We provide the first comprehensive evaluation of the effect of these codes on own structure survival and on neighbor spillovers via structure-to-structure fire spread. We then embed these empirical estimates in an economic model to calculate net social benefits of wildfire building codes as a function of local wildfire hazard and number of close neighbors. Finally, we use the model and data to compare the welfare effects of building codes, voluntary mitigation subsidies, and a no-policy alternative.

This analysis takes advantage of a new dataset that includes property-level data for the majority of US homes exposed to wildfire between 2003 and 2020, where "exposed" is defined as being included within the burn perimeter of a wildfire incident. We compiled the data by requesting post-incident damage censuses from numerous emergency management agencies and individual county assessors. We merged these lists of damaged homes to assessor data for the universe of (destroyed and surviving) homes inside wildfire burn areas. The data show that even during catastrophic wildfires, more than 50% of exposed homes survive unscathed (wildfire damage outcomes are typically bimodal, with homes either undamaged or totally destroyed). One of the key advantages of the new data is the ability

[5] Florida has construction standards for hurricane winds, and codes exist in various regions for winter storms and nonweather disasters, such as earthquakes and tsunamis (FEMA 2020). In flood-prone areas, US federal rules require homes to be elevated, and some localities have imposed even stricter requirements. California, Utah, Nevada, and Pennsylvania have statewide wildfire building standards, while in other states (notably Colorado), wildfire codes have been adopted at the local level (Insurance Institute for Business and Home Safety 2019). Australia, New Zealand, France, and Italy also have wildfire building codes (Intini et al. 2020).

to observe and learn from these surviving homes. The property-level loss information also distinguishes the wildfire data from floods and other disasters where loss data are typically available at the zip code or census tract level. In addition to the new loss data, the empirical work also leverages emerging tools in spatial analysis, including high-resolution aerial imagery and precise rooftop geocoding of structure locations.

The empirical design makes use of rich variation in building code requirements across space and over time. The complex nature of building regulation in California creates a patchwork of wildfire standards across localities. We also observe fires in other states that do not have wildfire building codes. In all of these places, we observe homes built before and after changes in California's codes. This identifying variation yields credible counterfactual predictions for how homes would have performed in the absence of California's standards. Our preferred statistical model is a fixed effects regression that compares the likelihood of survival for homes of different vintages located on the same residential street segment during the same wildfire event. These substreet fixed effects allow us to compare groups of homes that experience essentially identical wildfire exposures.

We find remarkable vintage effects for California homes subject to the state's wildfire standards. A 2008 or newer home is about 13 percentage points less likely to be destroyed than a 1990 home experiencing an identical wildfire exposure (a 34% reduction from the sample average of 39%). There is strong evidence that these effects are due to state and local building code changes: first after the deadly 1991 Oakland Firestorm and again with the strengthening of wildfire codes in 2008. The observed vintage effects are highly nonlinear, appearing immediately for homes built after building code changes. There are no similar effects in jurisdictions without wildfire building codes.

We also find that code-induced mitigation benefits neighboring homes, consistent with reduced structure-to-structure spread. These neighbor effects are in keeping with anecdotal reports of home-to-home spread as an important factor in urban conflagrations (Cohen 2000, 2010; Cohen and Stratton 2008).[6] Our results imply that, all else equal, the presence of a precode neighbor less than 10 meters away (within the distance fire experts refer to as the home ignition zone) increases a home's likelihood of destruction during a wildfire by about 1.7 percentage points (4.4%). This benefit is even larger when homes have multiple close neighbors.

We separately investigate whether the codes had an effect on housing market outcomes. Using data on home sales and construction near

[6] We are also aware of at least one insurance company that will not sell homeowners insurance to homes located next to a home with a wood roof in high-risk areas (Allstate Indemnity Company; California Department of Insurance 2018b).

borders of code-required areas across California, we show that neither home sale prices nor growth rates in code-required areas changed differentially in response to the building codes. This indicates that the welfare impacts of the building mandates are driven primarily by the costs and benefits of the mitigation requirement.

Finally, we develop an economic model that identifies the conditions under which mandated adaptation improves welfare. We use the model, our empirical results, and additional data on homes in wildfire hazard areas throughout California to understand (1) where universal mitigation investment is cost-effective and (2) the relative welfare properties of building codes, voluntary mitigation subsidies, and a no-policy alternative. With respect to the first point, our calculations find that wildfire building codes deliver unambiguously positive benefits in the most fire-prone areas of the state, especially where homes are clustered closely together and thus create large risk spillovers. In areas with more moderate wildfire risk, building standards for new homes can also be justified given reasonable assumptions about household risk aversion, the completeness of insurance, future increases in wildfire hazard, and/or cobenefits of building codes (such as reductions in public expenditures on wildland firefighting). On the other hand, the costs of retrofitting existing homes to meet current wildfire building standards are substantial, and our analysis suggests that full retrofits are economic only in areas with extreme wildfire hazard.

Turning to the policy comparison, we find that building codes with a degree of geographic targeting similar to those employed in practice increase welfare by about 20% relative to the no-policy alternative. We also find that welfare under building codes is only slightly lower than under an idealized perfectly differentiated mitigation subsidy. These favorable welfare results are driven by two features of disaster risk that have not been sufficiently appreciated in the literature. First, unlike many goods where regulators might apply bans or mandates, key determinants of the returns to mitigation investment are readily observable. This includes the level of wildfire risk and the number of close neighbors affected by risk spillovers. Second, the spatially continuous nature of disaster risk means that these factors are highly spatially correlated within neighborhoods. This combination of factors makes it feasible to target adaptation mandates to areas where almost all, if not all, homeowners will be made better off by investment.

These results are broadly relevant to natural disaster management. The standards-based approach that we study nearly halves loss risk when structures are exposed to the hazard. Moreover, our cost-benefit calculations imply that low take-up in the absence of standards is unlikely due to a lack of cost-effectiveness. These facts can inform policies to mitigate other risks—like floods, hurricanes, tornadoes, and heat waves—where

voluntary adaptation investment may also be limited by risk misperception and other potential market failures.[7]

This work also has immediate implications for wildfire policy. Our results imply that there are gains to be realized from strengthening building codes in other states and countries to match those in California and that a home's adherence to wildfire building standards has benefits not just for itself but also for its neighbors as well. This evidence is relevant to recent proposals in Colorado, Oregon, Washington, and other states (see, e.g., Profita 2021). Meanwhile, California is moving to expand the geographic coverage of designated wildfire hazard zones and reduce the ability of local jurisdictions to opt out of recommended standards. Separately, new California legislation from 2020 provides financial incentives for retrofits of existing homes in wildfire-prone areas.[8] The law specifically calls for support of cost-effective retrofits, a concept for which the evidence in this study is essential. Additionally, policymakers are confronting pressing issues of insurance rate reform in response to mounting wildfire losses. One key debate is the degree to which individual investments improve structure survival and should thus be rewarded through regulated insurance discounts (California Department of Insurance 2018a), such as California's Safer from Wildfires program. This paper's evidence on the effectiveness of such investments during real wildfires bears directly on this question.

Our work builds on previous studies of natural hazard mitigation. For wildfires, a number of engineering and forestry studies describe the effects of construction materials and vegetation management on structure resilience (Gibbons et al. 2012; Syphard et al. 2012; Syphard, Brennan, and Keeley 2014, 2017; Alexandre et al. 2016; Kramer et al. 2018; Syphard and Keeley 2019). Our paper focuses on the effects of a mandatory mitigation policy, while these previous studies measure the effectiveness of specific mitigation technologies (e.g., survival of homes whose owners did vs. did not choose to install fire-resistant exterior siding).

Engineering studies on building requirements for homes exposed to natural disasters typically calculate the value of building codes through modeling and simulation (e.g., FEMA 2020). Two empirical studies on the related topic of hurricanes do consider building codes, with conflicting

[7] Risk misperception is an important theme across disaster types: see Kunreuther and Michel-Kerjan (2011) for an overview of cognitive biases in disaster risk mitigation. Spillovers and other market failures differ across hazards. While direct physical risk spillovers are most pronounced for fires, other types of spillovers affect all disasters. Fu and Gregory (2019) document after-the-fact amenity spillovers from postflood relocation decisions, which presumably correlate with damage levels and thus ex ante mitigation. Large-scale public investments to manage flooding can also create longer-distance hydrologic externalities that may justify coordination of mitigation investments (Hummel et al. 2021; Bradt and Aldy 2023).

[8] See California's Senate Bill 63 (2021–22) and Assembly Bill 38 (2019–20).

results. Dehring and Halek (2013) is a small case study of several hundred homes during Hurricane Charley in 2004. Simmons, Czajkowski, and Done (2018) study aggregate zip code–level data on annual insurance claims by homes built in different decades to infer benefits of hurricane building codes in Florida. In contrast, our study uses highly granular property- and event-level loss data for a large sample of wildfires covering several states. In the flood context, Ostriker and Russo (2022) provide complementary evidence that building codes can reduce damages by forcing homeowners to build elsewhere or to elevate their homes. Their finding that targeting mandates toward high-risk areas improves social welfare echoes what we see in our setting. Finally, our work is methodologically related to a separate literature in economics on building codes and household energy consumption (Jacobsen and Kotchen 2013; Levinson 2016).

This study makes three main contributions. First, we provide the first comprehensive evaluation of the causal effects of wildfire building codes on own and neighboring structure survival. The causal estimates we document leverage a quasi-experimental empirical design, where previous work is primarily descriptive or relies on regression adjustment. The estimates of risk spillovers between neighbors make use of detailed information on the spatial relationships between nearby homes. Beyond the wildfire context, this result improves our understanding of disaster resilience under standards-based versus voluntary policies. Second, by embedding the empirical estimates in an economic model, we calculate the net social benefits of building codes and compare these with the benefits of alternative adaptation policies. Third, and finally, we compile and make available a comprehensive dataset of structure-level outcomes in wildfires across several states that, to our knowledge, is the most complete accounting in existence. This new dataset will enable future work on the economics of catastrophic wildfire risk.

The rest of the paper proceeds as follows. Section II discusses structure survival in wildfires and California's history of building code updates. Section III describes the data and spatial analysis. Section IV outlines the empirical strategy, and section V presents the results. Section VI develops the model of net social benefits of building codes and other policies, and section VII concludes.

II. Wildfire Building Codes in California and Other States

> Unlike a flash flood or an avalanche, in which a mass engulfs objects in its path, fire spreads because the requirements for combustion are satisfied at locations along the path.... A wildland fire cannot spread to homes unless the homes and their

> adjacent surroundings meet those combustion requirements. (Cohen 2000, 16–17)

Established forestry and engineering evidence supports the importance of the so-called home ignition zone in determining structure resilience to wildfires. The home ignition zone includes the home itself as well as an area extending 30 meters away from the structure. Fire scientists emphasize the elimination of flammable materials inside this zone (e.g., Cohen 2000, 2010; Calkin et al. 2014). This guidance applies to both vegetation around the home (defensible space) and the construction of the home itself, especially the roof.

Among US states, California has gone the furthest in mandating takeup of wildfire resilience investments by property owners.[9] However, the application of these codes varies throughout the state. In jurisdictions where the California Department of Forestry and Fire Protection (CAL FIRE) provides firefighting services (State Responsibility Areas [SRAs]), the state directly determines building standards. Within incorporated cities and other jurisdictions with their own fire departments (Local Responsibility Areas [LRAs]), local governments have historically had greater control over code requirements. The development of the modern standards began with the Oakland Hills Firestorm of 1991, which killed 25 people and caused US$1.5 billion in property damage. The tragedy led to a series of legislative actions during the mid-1990s that required more fire-resistant roofing and maintenance of vegetation immediately adjacent to the home. The first of these was the so-called Bates Bill of 1992 (Assembly Bill 337). Among other changes, the Bates Bill encouraged stronger building standards in LRAs by requiring CAL FIRE to produce maps of recommended Very High Fire Hazard Severity Zones (VHFHSZ). In LRAs, local governments could then choose whether to adopt these recommended hazard maps and thus the accompanying building standards. This designation process unfolded over several years, with hundreds of local governments adopting or rejecting CAL FIRE's proposed VHFHSZ maps at different times. According to Troy (2007), 151 of 208 local governments (73%) either adopted the VHFHSZ regulations or claimed to have promulgated equally strong rules.[10]

On the heels of the Bates Bill, Assembly Bills 2131 (1992) and 3819 (1994) increased requirements for ignition-resistant roofs. Roofing materials are rated class A, B, or C or are unrated.[11] Starting in 1995, the new

[9] Efforts to adopt statewide codes in Oregon, Colorado, and Washington are ongoing but continue to face substantial opposition (Sommer 2020; Booth 2023; SBCC 2024).

[10] For a detailed qualitative study of the determinants of local VHFHSZ adoption decisions, see Miller, Field, and Mach (2020).

[11] These ratings are earned through laboratory testing; e.g., the class A test involves placing a 12 × 12-inch burning brand on the roof material under high wind conditions for

codes required class B or C roofs on newly constructed or reroofed homes throughout California. In 1997, the requirement increased to class A roofs for high-hazard areas (a substantial improvement in fire resistance). However, the stronger class A standard applied only in SRAs and the subset of LRAs where local governments had adopted state-recommended VHFHSZ maps. Finally, Assembly Bill 423 in 1999 simplified enforcement of the new roofing codes by outlawing the use of unrated roofing materials throughout the state.

The collective effect of these mid-1990s building code reforms was to substantially increase the fire resistance of roofs on newly constructed homes. The roofing requirements also applied to existing homes but only at the time of roof replacement. Homeowners who replaced more than 50% of the roof surface in a single year were in principle obligated to comply.

California strengthened its wildfire codes again in 2008 with the so-called Chapter 7A standards of the California Building Code. These requirements apply to all homes built in 2008 or later in all SRAs and in LRAs where proposed VHFHSZ designations have been accepted. The codes apply to many dimensions of new homes. Roofs must be rated class A or B, eaves and exterior siding must be fire resistant, vents must covered by a fine wire mesh to resist ember intrusion, windows and doors must resist fire for at least 20 minutes, and decks and other building appendages must be built of noncombustible or otherwise fire-resistant materials. Chapter 7A also includes additional requirements for defensible space, which also applied to existing and new homes. However, in practice, the primary point of enforcement for these codes was at the time of new construction; enforcement effort for existing homes appears to have been limited (see e.g., Maclay 1997).

In summary, there were two key periods of change in California wildfire building codes: the mid-1990s and 2008. The strictness and uniformity of the codes varied across SRAs and LRAs. Codes were applied directly as written in SRAs, while local decision-makers exercised final control over the adoption of state-recommended codes in LRAs.

This study also considers wildfires in Arizona, Colorado, Oregon, and Washington. None of these had statewide wildfire building standards at the time of the included fires (Insurance Institute for Business and Home Safety 2019). Some local governments—particularly in Colorado—have

90 minutes. A roof's fire resistance depends on the roof covering material (shingles, tiles, concrete, etc.), the particular flame retardants with which that material may be treated, and the way that the roof components are assembled. Thus, there is not a one-to-one mapping between roof covering material and rating class, though untreated wood shingles are always unrated. For a detailed discussion of roof classifications, see Quarles et al. (2010) and ASTM International Standard E108, "Standard Test Methods for Fire Tests of Roof Coverings."

adopted local standards that include a diverse mix of rules about roofs, other construction materials, and/or defensible space. Our empirical analysis excludes a small number of fires in the comparison states that overlap jurisdictions with local wildfire building standards.[12] While the non-California homes in this study are not subject to mandatory standards, they—like the California homes—are targeted by a range of information and incentive programs that seek to increase voluntary home hardening. Programs active in these states include FireWise USA (National Fire Protection Association), the Community Wildfire Protection Plan program (US Forest Service and Department of Interior), the Fire Adapted Communities Coalition (numerous public agencies and nongovernmental organizations), the Ready, Set, Go! program (International Association of Fire Chiefs), and numerous other initiatives.

III. Data and Spatial Analysis

This section describes the construction of the database of wildfire damages, property tax assessment information, and the spatial analysis used to identify structure locations and distances to neighboring homes.

A. Homes and Damage Data

1. Damage Inspection Data

We sought to assemble as comprehensive a database as possible of administrative records for homes destroyed or damaged by wildfire in the United States. For recent wildfires in California, this information is managed by CAL FIRE. For earlier California fires and for fires in other states, we contacted individual county assessors (who track these damages in order to update property tax assessments) and other agencies to request historical records of structure damages. To our knowledge, the resulting database is the most complete accounting of US homes lost to wildfire.[13] In general and consistent with the case study observations in Cohen (2000), the vast majority of damaged structures are fully destroyed (i.e., partial damage outcomes are rare). We thus follow the literature and focus on destroyed as our primary outcome.

[12] These are the 2012 Waldo Canyon Fire, 2013 Black Forest Fire, and 2018 Mile Marker 117 Fire in El Paso County, Colorado (Quarles et al. 2013), and the 2012 High Park Fire and 2020 Cameron Peak Fire in Larimer County, Colorado (Larimer County 2020).

[13] Comparing with incident-level aggregate data in Incident Status Summary 209 (ICS-209) incident reports (St. Denis et al. 2023), we calculate that the wildfires in our data represent 70% of all of the wildfire-destroyed residential structures across the entire United States between 2003 and 2020 (the time period of our sample). Restricting to only the five states in our sample (California, Arizona, Colorado, Oregon, and Washington), we capture 85%. Restricting further to only incidents between 2010 and 2020, we capture 93%.

California 2013–20.—In California, the CAL FIRE Damage Inspection database is a census of destroyed and damaged homes following significant wildfire incidents during 2013–20. The data include street address and assessor parcel number, limited structure characteristics, and, for some fires, an additional sample of undamaged homes. The damage variable has four levels: destroyed (>50% damage), major (26%–50%), minor (10%–25%), and affected (1%–9%). Of these, destroyed is the most commonly reported damage category and the only category that appears consistently across all fires.

California 2003–13.—Data for pre-2013 wildfires in California come from two sources. For the 2003 and 2007 San Diego firestorms, we received damage assessment data from San Diego County. For other counties, CAL FIRE staff provided us with a large collection of unformatted historical damage assessment reports that we compiled and standardized to be usable for research.

Other states.—Using ICS-209 incident reports, we identified 15 counties in western states other than California with large numbers of structures lost to wildfire since 2010. We then contacted county assessors in each of these counties to request damage data. We received usable structure-level damage data from 10 of these 15 counties, including nine of the top 10 by structures destroyed.[14] Among these areas, we conduct an extensive search to rule out locations where wildfire building codes were in place prior to the relevant wildfire incident.[15]

2. Property Tax Assessment Data

We merge the damage records to comprehensive assessment data for all US homes from the Zillow Transaction and Assessment Database (ZTRAX). The ZTRAX data include information on year built, effective year built (in the case of remodels), building square footage, and other property characteristics. The merge from damage data to ZTRAX uses assessor parcel numbers, and we validate the accuracy of this merge by comparing street addresses across the two datasets. We restrict the data to include only single-family homes, which account for most properties inside the wildfire perimeters in our sample.[16] For each incident, we merge the

[14] Specifically, homes in the following non-California counties are represented in our data (in order of structures destroyed): Jackson (Oregon), Okanogan (Washington), Marion (Oregon), Lane (Oregon), Grand (Colorado), Lincoln (Oregon), Jefferson (Oregon), and Yavapai (Arizona).

[15] We document that process in app. sec. A.7 (apps. A–E are available online). As a result, we do not include homes located in El Paso County and Larimer County, since they both had county-wide wildfire building codes prior to the wildfire incidents we study.

[16] In the CAL FIRE Damage Inspection data, which include comprehensive information on structure category, single residences represent 91% of all destroyed structures (excluding minor structures, such as sheds). Of all single residences, single-family homes represent 82%

damage data to the most recent historical assessment data from the prefire period (assessments are conducted annually or every other year in nearly all counties). In other words, we merge to the population of single-family homes that existed immediately prior to the start of the fire. Table A5 (tables A1–D6 are available online) documents the full list of wildfire incidents in the dataset, the number of single-family homes inside of each perimeter, and the share destroyed.

B. Spatial Analysis and Dataset Construction

1. Identifying Structure Rooftop Locations

This study uses the physical locations of the homes in the data in two ways. First, homes must be spatially assigned to building code jurisdictions and to wildfire burned areas. Second, the measurement of spillovers across properties requires precise distances between neighboring structures. The street address–based geocoding methods typically used in academic research are not sufficiently detailed for this second purpose, which requires accurate structure locations at a meter scale. We solved this challenge by combining several spatial datasets to identify precise rooftop locations. First, we limit the population of ZTRAX homes to all homes in zip codes where at least one home was destroyed. We then merge these ZTRAX records to parcel boundary maps from county assessors using assessor parcel numbers. This yields a parcel polygon for each home. We then use comprehensive building footprint maps from Microsoft to identify the largest structure overlaying each parcel.[17] We call this the footprint location. Figure 1 shows an example for Redding, California, in the area of the 2018 Carr Fire. Gray lines are parcel boundaries from the Shasta County Assessor. Polygons are building footprints. Squares and circles show the assigned rooftop locations for each structure. Circles show homes that are reported as destroyed in the damage data.

This rooftop geocoding method generates highly accurate locations, but it is dependent on the availability of high-quality parcel boundary geographic information system (GIS) data. In areas where such data are not available (representing 7% of homes in the final analysis dataset), we instead geocode home locations using the ESRI StreetMap Premium geolocator, a commercially available address-based product. Our quality checking shows that these locations (henceforth, "address-based locations")

of the data. The remainder are primarily manufactured or mobile homes, which are not subject to California's wildfire building codes.

[17] See https://github.com/microsoft/USBuildingFootprints for the Microsoft US Building Footprints Database.



FIG. 1.—Building and validating dataset. The figure shows home roof locations, damage indicators, and distance between structures. *A*, Homes affected by Carr Fire (2018). Markers are geocoded structure locations. Circles indicate structures reported as destroyed in the damage inspection data. Squares indicate all other homes in the data. The background image is aerial imagery before and after the Carr Fire from NearMap. Building shapes and parcel outlines are the building footprint data and assessor parcel boundary data used to identify structure locations. *B*, Examples of calculated distances between structure walls. Images are prefire aerial imagery of homes affected by the Thomas Fire (2017) and Tubbs Fire (2017). The figure shows the wall-to-wall distance from the structure marked 0 to the other homes.

are generally reliable to the parcel level but not always to the structure rooftop level. Appendix section A.5 describes the geocoding in more detail.

2. Validating Locations and Damage Reports

We quality check the calculated property locations and the damage report data using high-resolution aerial imagery from NearMap. The base image

in figure 1 shows an example. The detailed imagery allows us to manually confirm the accuracy of structure locations, which closely coincide with the building footprints in the figure. In addition, the NearMap imagery includes postfire surveys for many of the incidents in our database. Figure 1 illustrates how destroyed properties are readily visible in these surveys, which allows us to confirm the accuracy and completeness of the damage data. Table A3 reports accuracy rates in a random sample of homes. For damage reports, 99% of reported outcomes match the ground truth imagery. For rooftop locations, 98% of the assigned structure locations are on top of the structure rooftop in the ground truth imagery (with 99%+ accuracy in densely developed areas). Locations that rely on street address–based geocoding tended to be accurate to the parcel but not always to the actual structure rooftop: about 75% of these assigned locations are on top of the structure rooftop in the ground truth imagery.

3. Spatial Merge to Wildfire Perimeters, Code Jurisdictions, and Topography

We restrict the dataset to homes located within final wildfire perimeters (plus a 10-meter buffer). Depending on the state and time period, these digital perimeter maps come from the California Forest and Range Assessment Program, the Monitoring Trends in Burn Severity dataset, or the National Interagency Fire Center.

We merge the homes data to spatial data on fire protection responsibility (SRA vs. LRA) and designated fire hazard severity zone (FHSZ) that together determine building codes in a given location in California. We use historical GIS maps provided by CAL FIRE to assign homes to code regimes according to the codes in effect when the home was built.[18] Section IV.A discusses the assignment of homes to code regimes in more detail.

For each home, we also compute measures of topography and wildfire hazard: we obtain ground slope, elevation, and vegetation type from 30-meter resolution grids provided by LANDFIRE (Rollins 2009), and we compute a measure of wildfire hazard using data from the US Forest Service Wildfire Risk to Communities project (Scott et al. 2020). We refer to this measure as wildfire hazard. It represents the annual probability of moderate to severe wildfire exposure for each home and is calculated as the product of burn probability (the total annual wildfire probability) and flame length exceedance probability 4 (conditional on any fire, the probability that the fire will reach moderate or greater threat status). This measure represents a snapshot of topography and wildfire risk from 2013

[18] For SRA/LRA boundaries, the historical map data include updates in 1990, 1996, 2003, 2005, and annually from 2010 to 2020. For FHSZs, the historical map data include updates in 1985, 1998, 2007, and 2008.

and 2014, prior to most of the wildfires experienced by homes in the dataset.

4. Calculating Distances between Neighboring Homes

We construct two measures of distance between homes. The first is the minimum distance between the building footprint polygons associated with the two structures (henceforth, "wall-to-wall" distance). This measure is available only for homes where we assign locations based on building footprints. The second metric uses the distance between assigned point locations, which are available for all homes in the dataset. We call this metric the centroid-to-centroid distance because these points are meant to correspond to the center of the roof.[19] The wall-to-wall distance is our preferred measure because it more accurately captures space between homes and because the footprint-geocoded locations are more accurate than the address-based location points (table A3). Our main estimates of neighbor spillovers use the restricted sample of homes for which wall-to-wall distances are available. For robustness, we also show specifications that use centroid-to-centroid distances and the full sample of homes.

We identify potentially at-risk neighbors for each home in the final dataset. We define potentially at-risk neighbors as those within 200 meters (using centroid-to-centroid distances) of the focal home and included the given fire perimeter (plus the buffer). Figure 1*B* shows two examples. Each image shows wall-to-wall distances (in meters) from the structure marked 0. Figure A3 (figs. A1–D2 are available online) summarizes the distribution of number of neighbors at various distances.

5. Data Summary

The final dataset includes 45,093 single-family homes exposed to 104 wildfires in California, Arizona, Colorado, Oregon, and Washington between 2003 and 2020. This sample includes only homes for which the key variables are observed.[20] Thirty-nine percent of these were destroyed. Table A1 provides descriptive statistics. Figure A1 shows the distribution of year built and average share destroyed by year built for the full dataset.

[19] When using centroid-to-centroid distances, we add 20 meters to the measured distance to ensure comparability with the wall-to-wall distances, since 30 meters in centroid-to-centroid distance roughly corresponds to 10 meters in wall-to-wall distance. The median building footprint area in the sample is 260 meters^2. A hypothetical circular roof would thus have a radius of 9.1 meters, and the centroid-to-centroid distance between two such homes would be 18.2 + wall-to-wall distance.

[20] The key variables are those we use in our preferred specification, eq. (1): whether the home was destroyed, vintage, jurisdiction, period (year built), ground slope, and location.

IV. Empirical Strategy

This section describes the empirical design used to measure the effect of wildfire building codes on structure survival. To fix ideas, figure 2 provides an example of the merged dataset for the 2018 Woolsey Fire in Los Angeles County. Plus signs indicate locations of surviving and destroyed single-family homes inside the final fire perimeter. The street map data give a sense of development density. The intensity of losses varies significantly within the burned area. Near Malibu, a large share of affected homes were lost. Further north, however, there are several areas where most homes inside the fire perimeter escaped destruction. These

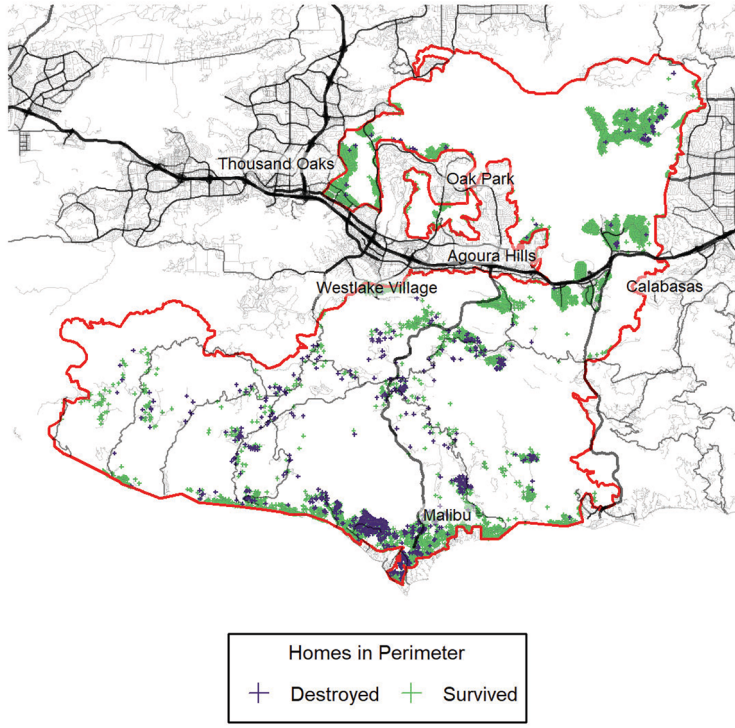


FIG. 2.—Structure-level outcomes in Woolsey Fire. The figure shows single-family homes in Los Angeles and Ventura Counties that were included within the Woolsey Fire perimeter in 2018. The map is an example of the merged inspection, assessor, and fire perimeter data for one fire in our dataset. Plus signs indicate the locations of all single-family homes included in the assessment data that were located inside the final Woolsey Fire perimeter. Homes with dark shading are reported destroyed in damage inspection data; homes with light shading are all remaining homes in the assessment data. Blank areas within the fire perimeter indicate that single-family homes were not present at the time of the fire; these are mostly mountainous state park and national recreation areas. Street map data are from Open Street Map.

differences reflect varying fire conditions, firefighter response times, landscape vulnerability, structure characteristics, and potentially numerous other factors. This heterogeneity adds noise to empirical analysis of structure survival. It may also introduce bias if year built or other structure traits vary similarly within burned areas. We address these challenges using an empirical design that compares the likelihood of survival for homes of different vintages on the same residential street during the same wildfire. We attribute these vintage effects to building codes by comparing vintage effects across jurisdictions with and without wildfire building codes.

A. Treatment Groups

Our empirical design focuses on homes in three types of jurisdictions. The first are homes in SRAs, where compliance with California building codes was mandatory and fully determined by maps released by CAL FIRE. The second are homes within LRAs that were ever recommended by CAL FIRE as VHFHSZs (henceforth, "LRA-VHFHSZs"). This group includes all proposed local VHFHSZs regardless of whether local governments accepted the designation. There is no centralized database that records local VHFHSZ adoption decisions, but Troy (2007) reports high rates of adoption.[21] In principle, however, it is possible that the maps used by LRAs to determine building code requirements could vary slightly from the original recommendation maps provided by CAL FIRE. For that reason, we consider the LRA-VHFHSZ homes as an intention-to-treat treatment group.

The third group of homes are those located in jurisdictions without wildfire building codes (henceforth, "no-codes") that were included in wildfire incidents during our sample period. These jurisdictions are locations in Arizona, Colorado, Oregon, and Washington without any state or local wildfire building codes. As discussed in section III, we are able to collect data for 10 of the 15 counties with the most destroyed homes outside of California, so this group includes the majority of available observations within the universe of possible comparison homes.[22] Panel B of table A1 reports the number of homes in each of the treatment groups.

[21] In addition, historical news accounts show that cities that rejected the official VHFHSZ designation often still adopted the underlying code requirements in the recommended areas. This seems to have been an attempt to achieve the state-recommended resilience requirements while avoiding the VHFHSZ label because of fears about its effect on property values (Snyder 1995; Stewart 1995; Sullivan 1995; Grad 1996; Yost 1996). According to one state fire official's response, "We didn't care if they called it a nuclear-free zone, as long as they adopted the regulations" (Maclay 1997).

[22] A small number of homes built in parts of California that were not included in California's wildfire building code zones have also experienced wildfires. We use this group in sensitivity checks as an alternative comparison group. Our preferred specifications focus

B. Identification

The empirical design takes advantage of variation across home vintages and code jurisdictions. We first estimate the difference in likelihood of destruction when confronted with a wildfire (henceforth, "wildfire resilience") for pre- and postcode homes in regulated regions. To isolate the effect of construction practices as opposed to other factors that might vary with home vintage (such as the intensity of the wildfire upon reaching a given neighborhood), our preferred specification includes spatially granular fixed effects that compare nearby homes on the same residential street (substreet fixed effects) during the same wildfire incident. Intuitively, this research design treats every substreet as its own natural experiment, where all of the homes in each substreet are on similar terrain and are equally exposed to the same wildfire but—for homes in code-required jurisdictions—face different building requirements depending on whether they were constructed before or after the codes were in place.

These localized comparisons rule out bias from omitted variables that might differ across neighborhoods or communities. As one example, this substreet design neutralizes potential differences in firefighting effort due to characteristics of the fire-threatened community, as discussed in Plantinga, Walsh, and Wibbenmeyer (2022) and Baylis and Boomhower (2023), since these differences are determined by average characteristics over a broad area.

We then ask whether the measured improvements in wildfire resilience for regulated California homes are the result of the state's building codes. The relevant identification assumption is that no other important determinants of wildfire resilience changed at the same time as the California building code changes that we observe. To support this assumption, we marshal several lines of evidence. First, we use an event study analysis to examine whether the time pattern of observed improvements in California lines up with changes in building code policies. Second, we consider the evolution of wildfire resilience for homes built over the same time periods in other states without wildfire building codes. This additional evidence—which makes use of a novel West-wide wildfire outcomes dataset that we compiled for this purpose—speaks to the question of whether something else about the way US homes were built changed exactly at the same time that California changed its wildfire building codes. In appendix section B.1.1, we also consider the evolution of wildfire resilience

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on the non-California comparison homes for two reasons. First, the within-California comparison areas contain relatively few recently built homes. Second, the ability of local governments to amend the proposed VHFHSZ maps prior to adoption may have led to some municipalities requiring to-code construction in areas not included in the proposed VHFHSZ designations from CAL FIRE, such that the treatment status of these comparison homes may be somewhat ambiguous to the researcher.

for homes built in parts of California that were never subject to California's wildfire building codes. This final line of evidence speaks to the question of whether something else changed about the way that homes were built only in California at the same time that California introduced its building codes.

C. *Own Structure Survival*

1. Flexible Vintage Effects

We begin the regression analysis with the following binned regression model for home i on street s exposed to wildfire incident f. We estimate this model separately for the SRA, LRA-VHFHSZ, and no-codes groups:

$$\text{Destroyed}_{isf} = \sum_{v=v_0}^{V} \beta_v \text{Vintage}_i^v + \gamma_{sf} + X_i\alpha + \epsilon_{isf}. \quad (1)$$

The outcome Destroyed_{isf} is an indicator variable equal to 1 for destroyed homes and 0 otherwise. Vintage_i^v represents indicator variables equal to 1 if house i's year built falls into bin v. The vintage bins vary in size depending on the availability of data in a given jurisdiction, and the omitted bin is the vintages just prior to the implementation of the codes. The main parameters of interest are the coefficients β that correspond to these vintage bins. These give the effect of each vintage on probability of survival when exposed to wildfire. In our preferred specification, we use substreet fixed effects γ_{sf}, which include separate indicator variables for groups of 25 homes on the same street within fire perimeter f. We interact the substreet fixed effects with jurisdiction indicators to account for the small number of incidents that cross jurisdictional boundaries. We also estimate models with incident fixed effects (which compare homes inside the same wildfire perimeter) and models with street fixed effects (which compare groups of homes on the same street).

The additional control variables X_i include controls for wildfire vulnerability at the home site. These include, at minimum, ground slope and vegetation type. For some alternative specifications, we also include elevation and property characteristics (lot size, building square footage, number of bedrooms), though this reduces the number of observations used in the analysis since these variables are not perfectly observed throughout our sample.

2. Pooled Regression

We summarize the overall effects of the wildfire building standards using a regression model that pools jurisdictions and vintages. We divide the sample into three vintage categories indexed by v: homes built before

1998 (the omitted category), between 1998 and 2007, and in 2008 and onward. The latter two periods correspond to the end of the mid-1990s roofing reforms and the introduction of the Chapter 7A requirements. Jurisdictions are indicated by Jurisdiction_i^j, which captures whether the home was built in the SRA, an LRA-VHFHSZ, or a no-codes jurisdiction:

$$\text{Destroyed}_{isf} = \sum_{v=v_0}^{V}\sum_{j=j_0}^{J}\beta_{vj}\text{Vintage}_i^v \times \text{Jurisdiction}_i^j + \gamma_{sf} + X_i\alpha + \epsilon_{ijf}. \quad (2)$$

The six β_{vj} estimated in this regression are the coefficients of interest, and they capture the change in destruction probability between the pre-1998 homes in each jurisdiction and homes built in later years. This specification emphasizes temporal changes within jurisdiction and is directly comparable to the finer bin estimates provided by equation (1).

D. Structure-to-Structure Spread

To measure the effect of code-driven mitigation on likelihood of structure-to-structure spread in code-required jurisdictions, we also estimate the effect of building vintage on likelihood of survival for neighboring homes in SRAs and LRA-VHFHSZs:

$$\text{Destroyed}_{isf} = \sum_{d=1}^{D}\rho_d\text{Precode}_{id} + \sum_{d=1}^{D}\phi_d\text{Postcode}_{id} + \gamma_{sf} + X_i\alpha + \epsilon_{isf}. \quad (3)$$

As in equation (1), this specification controls for own year of construction and substreet fixed effects. The additional regressors Precode_{id} and Postcode_{id} are the number of neighbors within various distance bins d that were built before and after wildfire building codes for each home i. Homes are considered postcode in 1998 in the SRA and in the year the jurisdiction was first recommended as a VHFHSZ in LRA-VHFHSZs. The coefficients ρ_d and ϕ_d for $d = 1, \ldots, D$ give the effect of these neighbors on own structure survival. Our preferred specification uses 10-meter bins of wall-to-wall distance. The sample is restricted to California homes (since we can reliably calculate exact home locations for only California homes) built in code-required jurisdictions (since there are too few homes in no-code jurisdictions within California to estimate neighbor effects by distance bin). We further drop condominiums and townhomes to focus on detached single-family homes.

This regression identifies the causal effect of code-induced mitigation by neighboring homes if the code regime for neighboring homes is uncorrelated with other determinants of structure- and neighborhood-level risk. This assumption is bolstered by the substreet fixed effects, which focus on highly local variation, and by comparing between homes with pre- or postcode neighbors immediately adjacent versus those with neighbors

farther away. The motivation behind this approach is that even within these narrow comparisons and even after controlling for own age, the age of a home's neighbors may still be correlated with other wildfire risk factors. Properties located 50–100 meters away are outside of the 30-meter home ignition zone and so present more limited direct ignition threat but should otherwise be subject to the same potential omitted variables as directly adjacent homes.

V. Results

A. Own Structure Survival

1. Graphical Evidence

Figure 3 shows the average rate of destruction for SRA homes in our sample according to year of construction. About 35% of exposed homes built prior to the mid-1990s were destroyed during the fires in our sample. These destruction probabilities begin to fall for homes built after the mid-1990s, decreasing quickly to about 20%. This sharp improvement in resilience corresponds in time to the post–Oakland Firestorm building reforms. Figure A2 shows that homes built before and after the building code changes are otherwise comparable. The introduction of building codes does not coincide with changes in site-level predictors of fire risk, like ground slope, or in structure characteristics such as building square footage.

There is also some evidence in figure 3 that homes built before about 1980 may be less likely to be destroyed than homes built just prior to the roof requirements. This could reflect the fact that these older homes are more likely to have been reroofed at least once after the mid-1990s and complied with the requirement for ignition-resistant materials at roof replacement. This pattern would imply a replacement cycle of about 30–40 years. Actual data on roof service lifetimes are scarce, but this period is within the range proposed by the National Association of Home Builders (2007) and other sources. To the extent that some pre–building code homes may be reroofed with code-compliant materials, our estimates of building code effects are conservative.

Figure 4 presents the vintage effect estimates from equation (1). Figure 4*A* shows homes in the SRA, where Wildland-Urban Interface (WUI) building codes are mandatory. The markers show estimates and 95% confidence intervals for 2-year vintage bins. The omitted bin is 1987–88, so these estimates can be interpreted as percentage point differences in likelihood of destruction relative to a 1987 home. The vintage effects are flat prior to about 1993 and then begin to decrease sharply during the 1995–99 period. The point estimates suggest additional reductions in loss probability following the adoption of the Chapter 7A codes in

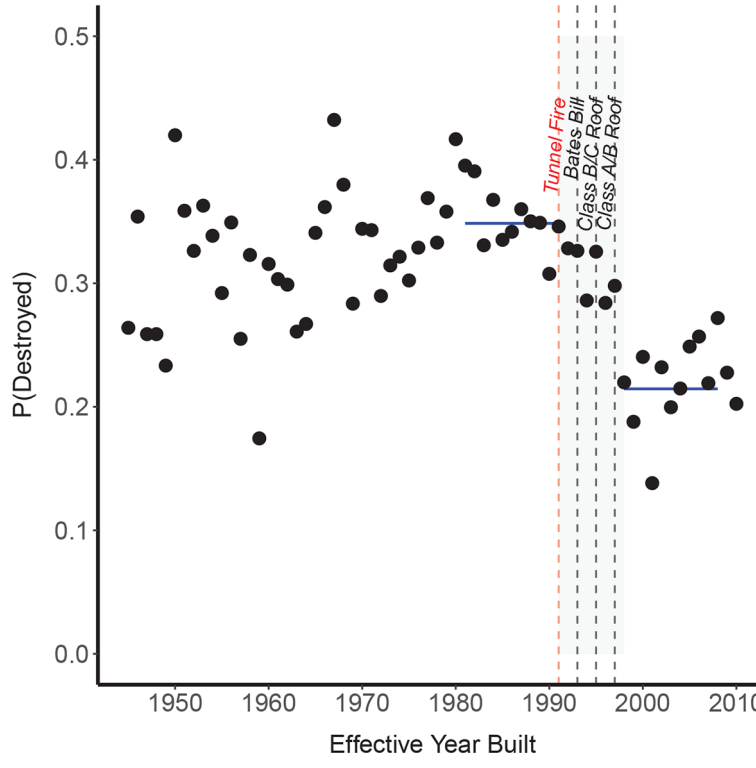


FIG. 3.—Share destroyed by year built in mandatory code jurisdictions. The figure shows the share of homes inside wildfire perimeters that were destroyed according to the year that the home was built. The sample is limited to homes in SRAs. The two horizontal lines show 10-year averages.

2000. This additional reduction could reflect the increased stringency of the revised structural requirements in the Chapter 7A reforms and the introduction of compliance with vegetation management regulations, although the small number of homes in those bins leads to somewhat noisy vintage estimates. The overall difference in loss probability between a 1987 home and a post-2008 home is about 15 percentage points.

Figure 4*B* shows homes in LRAs that CAL FIRE recommended for VHFHSZ designation. These jurisdictions again show flat trends in resilience prior to the 1991 Oakland Firestorm and subsequent Bates Bill. Figure 4*B* shows that after the Bates Bill took effect, steady improvements persisted for about 12 years. The slope of these improvements appears more gradual than in SRAs, which is consistent with varied timing of adoption of the recommended codes across hundreds of individual municipalities. The post-2008 estimates are again noisy but imply further

improvements in resilience following adoption of the Chapter 7A building codes.

Finally, figure 4*C* shows vintage effects for homes in jurisdictions not subject to California's codes. This includes fires in Arizona, Colorado, Oregon, and Washington with no state or local wildfire building codes. There are relatively few homes in this group, so we use wider 10-year vintage bins to increase precision. Unlike figure 4*A* and 4*B*, there is little evidence of improved resilience for homes built since the mid-1990s in these jurisdictions.

2. Regression Estimates

Table 1 summarizes the effects of wildfire building codes on structure resilience. We report separate estimates for the SRA, LRA-VHFHSZ, and no-codes groups. The various group × time period estimates can be interpreted as percentage point differences in likelihood of destruction relative to a pre-1998 home in the same jurisdiction.[23] Column 1 shows a specification with incident fixed effects. These incident dummies absorb fire-specific differences in severity and preparedness. In this specification, SRA homes built during 1998–2007 or 2008–16 perform 12.6 and 15.8 percentage points better, respectively. A similar pattern exists for LRA-VHFHSZs, with improvements of 6.7 percentage points for 1998–2007 homes and 10.7 percentage points for 2008–16 homes. Except for 1998–2007 LRA-VHFHSZ homes, all of these estimates are statistically distinguishable from zero at conventional levels. Similar improvements are not apparent in the no-codes comparison group, where homes built in the latter two vintage categories do not experience statistically significant changes in destruction rates. This is further evidence that observed improvements in the regulated jurisdictions are due to building codes as opposed to other time-varying factors. The regression also includes controls for topography and vegetation. As expected, slope steepness at the home site increases vulnerability. A home on a 10° slope would be 5 percentage points less likely to survive than an otherwise identical home on flat ground.

Columns 2–4 of table 1 show specifications with various street × incident fixed effects. These more granular fixed effects adjust for potential differences in fire intensity or home characteristics within the burned area. Column 2 reports estimates from a specification that controls for street fixed effects, identifying differences in destruction rates across homes of different vintages but on the same street. Again, we find that homes built in SRA and LRA-VHFHSZ jurisdictions after the California building code mandates came into effect experience large declines in

[23] Note that group-specific means are absorbed by the fixed effects in each specification.

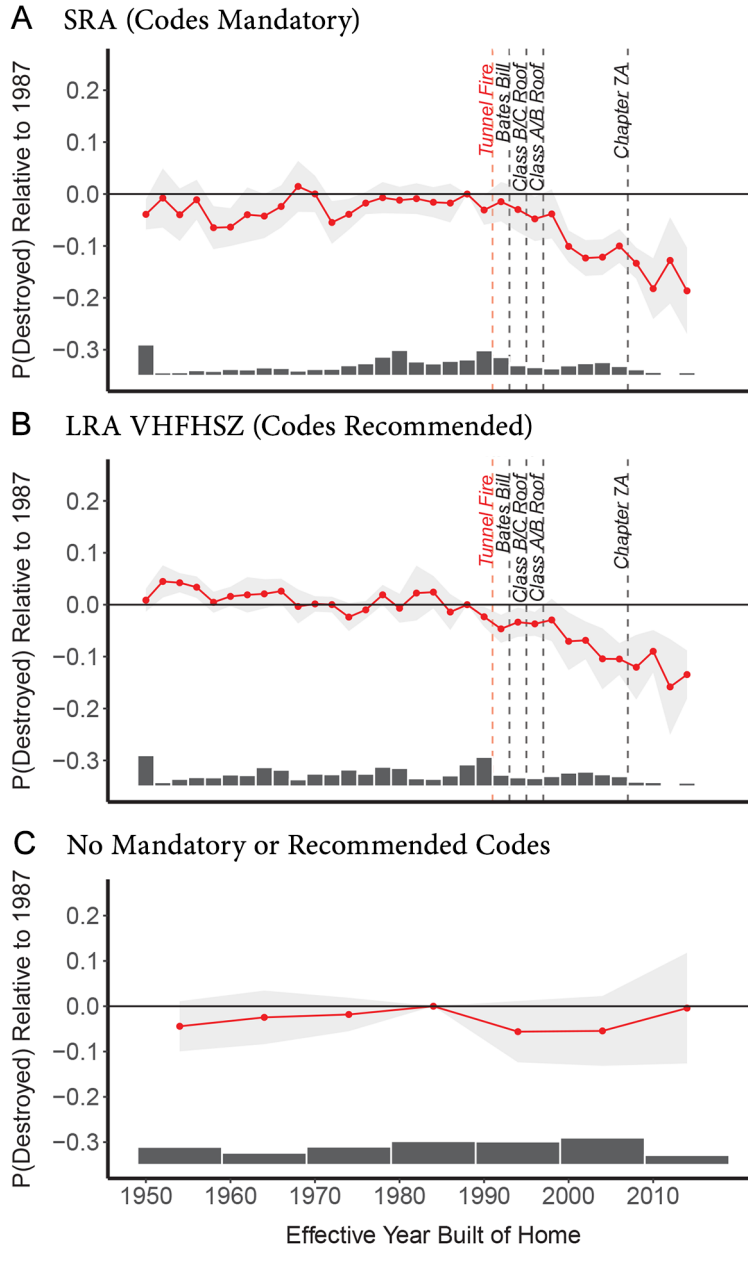


FIG. 4.—Estimated vintage effects by building code jurisdiction. The figure plots point estimates and 95% confidence intervals from event study regressions of home destruction on year built, split by jurisdiction. The outcome variable is an indicator for Destroyed, and the bin indicators group homes into either 2- or 10-year bins of effective year built. Each regression

TABLE 1
BUILDING CODE EFFECTS ON OWN DESTRUCTION

	Fixed Effects			
	Incident	Street		
	(1)	(2)	(3)	(4)
1998–2007 × SRA	−.126***	−.088***	−.089***	−.091***
	(.021)	(.009)	(.008)	(.010)
2008–16 × SRA	−.158***	−.133***	−.131***	−.139***
	(.032)	(.021)	(.019)	(.021)
1998–2007 × LRA-VHFHSZ	−.067	−.082***	−.071***	−.070**
	(.042)	(.028)	(.023)	(.027)
2008–16 × LRA-VHFHSZ	−.107*	−.122***	−.111***	−.109***
	(.060)	(.035)	(.026)	(.029)
1998–2007 × no-codes	.0005	−.045	−.038	−.029
	(.056)	(.031)	(.025)	(.027)
2008–16 × no-codes	−.028	.004	.005	.032
	(.046)	(.039)	(.044)	(.050)
Ground slope (°)	.004***	.005***	.004***	.004***
	(.001)	(.0007)	(.0007)	(.0006)
Fuel model indicators	Yes	Yes	Yes	Yes
Additional controls				Yes
Incident fixed effects	Yes			
Street fixed effects		Yes		
Substreet fixed effects			Yes	Yes
Observations	45,093	45,093	45,093	36,176
R^2	.39	.62	.66	.67
Dependent variable mean	.39	.39	.39	.44

NOTE.—The table shows estimates and standard errors from regressions of home destruction on vintage × jurisdiction indicators. The outcome variable is an indicator for Destroyed. The included home vintage categories represent homes with effective year built in 1998–2007 and 2008–16, and the excluded category is homes with effective year built before 1998. SRA and LRA-VHFHSZ are treatment groups where wildfire building codes applied after 1998, while no-codes is the comparison group of homes in other states without building codes. Incident fixed effects are dummies for each wildfire. Street fixed effects include separate dummies for each street × incident. Substreet fixed effects use separate dummies for each group of 25 adjacent homes on the same street × incident. Fuel model indicators are dummies for Anderson fire behavior fuel models. Additional controls include lot size, home square footage, number of bedrooms, elevation in meters, and wildfire hazard. Table A6 reports all estimated coefficients. Standard errors are clustered by incident.

* $p < .10$.
** $p < .05$.
*** $p < .01$.

destruction, while non-California homes show no change over the same period.

Column 3 reports the preferred specification using substreet fixed effects: here we examine differences in destruction rate for homes of different

includes substreet × incident fixed effects and other controls described in the text. *A* shows homes in the SRA and uses 2-year bins. *B* shows homes in state-recommended LRA-VHFHSZs and uses 2-year bins. *C* shows homes in no-codes jurisdictions in Arizona, Colorado, Oregon, and Washington and uses 10-year bins. Standard errors are clustered by incident. The histogram below each panel shows the relative number of observations in each bin.

vintages within sets of 25 homes on the same street. We find that SRA homes experience declines of 8.9 percentage points (on a mean of 39 percentage points) if they were built between 1998 and 2007 and 13.1 percentage points if they were built between 2008 and 2016. LRA homes experience analogous declines of 7.1 and 11.1 percentage points. By contrast, homes outside of California do not experience statistically significant changes in destruction rate. The R^2 with substreet fixed effects is notably higher than with incident fixed effects (0.66 vs. 0.39), implying that these fixed effects remove variation in fire severity and other factors within incidents that might otherwise threaten identification.

Even with this rich set of fixed effects, differential trends in home construction across space could lead to either home or location characteristics shifting differentially across locations over the period we study. Table A7 tests for meaningful changes in the covariates using the standardized mean difference approach described in Imbens and Rubin (2015). We do not see meaningful differential changes in slope, elevation, lot size, bedrooms, or wildfire hazard. However, homes in code-required jurisdictions have increased in square footage slightly more during this time period than homes in jurisdictions where there are no codes in place. If larger homes—having a large surface area for ignition—are more likely to burn in wildfires, then this imbalance would cause the effects of mandatory building codes we document in columns 1–3 to be biased toward zero (i.e., our estimates would be conservative).

To account for any covariate imbalances directly, column 4 adds lot size, number of bedrooms, home square footage, lot elevation, and wildfire hazard as additional controls. These characteristics are missing for about 20% of homes, which shrinks the sample in column 4. Nevertheless, the point estimates increase slightly in magnitude relative to column 3 for homes in areas with codes and remain statistically insignificant and close to zero for areas without them. Table A6 reports all estimated coefficients for the specifications.

3. Additional Results and Sensitivity Checks

Appendix B documents an extensive series of additional results and sensitivity checks on the main findings. We summarize these checks here and reserve complete descriptions for appendix B. In general, the findings are insensitive to alternative choices of estimation samples, included control variables, and estimation approaches.

Alternative estimation samples.—Appendix section B.1.1 considers several sensitivity checks on the main estimates that use different sets of homes as the estimation sample. First, we report specifications using the small number of California homes that were not subject to building codes as comparison homes. Second, we estimate a specification that includes

only homes built before 2015 and fires occurring after 2015, and we drop homes that are included in more than one fire perimeter. Across these alternatives estimation choices, the qualitative findings remain the same: postcode homes in jurisdictions where codes were required survive at considerably higher rates than precode homes, while the estimates for homes in the no-codes group remain close to zero.

Differential firefighting effort.—We consider whether targeted firefighting effort could explain part of the resilience effects that we measure. As discussed previously, because our empirical design compares outcomes for homes on the same substreet, it would need to be the case that suppression effort is targeted not only broadly across threatened neighborhoods but also within groups of nearby homes on the same street. Columns 6 and 7 of table B1 test for this possibility by restricting the sample to only homes in very large wildfires, where the ability of firefighters to protect individual homes is extremely limited.[24] We find that the estimated resilience improvements from building codes are similar when we include only homes involved in fires that threatened more than 1,000 homes or where fire managers reported severe personnel constraints. In the spirit of completeness, table B1 also shows that the results are unaffected by dropping homes located less than 50 meters inside the edge of the final wildfire perimeter, which is another way of testing for highly localized targeting of firefighting effort.

External validity.—Potential community-scale differences in firefighting effort and other factors could in principle raise questions of external validity. Is the sample of homes that actually experience wildfires representative of the broader population of homes at risk of wildfire? To help answer this question, in table A2 we compare characteristics of homes that experienced a wildfire to other sets of potentially at-risk homes. We find that the main estimation sample is broadly similar on observables to three increasingly restrictive comparison groups: all California homes in census tracts with average wildfire hazard exceeding 0.1% (the sample we consider in sec. VI), homes in areas subject to California's wildfire building codes, and homes in SRA jurisdictions. We interpret this as evidence that our findings are likely to be broadly applicable to homes in other at-risk areas in California.

Homes contributing to estimation with fixed effects.—To better understand which homes are contributing to the estimation, in appendix section B.1.2 we document how many homes are on streets or substreets with variation in exposure to building code regulations (and therefore contribute to identification with street and substreet fixed effects). In brief, around half of the homes in our sample are on streets with both pre- and postcode

[24] For example, the 2018 Camp Fire in Paradise, California, was burning more than 80 acres per minute at one point (Simon 2018).

homes. The subsample with within-street variation is not substantially different on observables.

Attenuation due to spillovers.—Appendix section B.1.3 considers the potential for attenuated estimates of own home resilience due to the spillover effects in section V.B. We implement a regression that controls directly for the number of pre- and postcode near neighbors in the street fixed effects regression. Ultimately, the differences in the estimated building code effects across these approaches—substreet fixed effects, incident fixed effects, and substreet fixed effects directly controlling for spillovers—are small.

Alternative statistical specifications.—Appendix section B.1.4 presents a set of sensitivity tests that use the original estimation but vary the statistical specification used by introducing increasingly granular sets of fixed effects and alternative research designs. The variations on our preferred model include a specification that includes only fixed effects as controls, substreet fixed effects that also account for side of street, propensity score matching, formal difference-in-differences, and an analogous logit model. For all of these specifications, we find quantitative estimates virtually identical to those shown above, and in many cases the point estimates are larger in magnitude than those we document here.

Neighborhood estimates.—Appendix section B.3 describes and estimates a model that regresses average neighborhood-level outcomes on proportions of homes in that neighborhood built to code. We find slightly larger average effects when the model is estimated in this way, which is consistent with the main resilience estimates presented in table 1 and suggestive of the additional within-neighborhood spillover benefits we consider in section V.B.

B. Spillovers to Neighboring Properties

This section discusses the spillover benefits of code-induced mitigation to neighboring homes. Figure 5 shows regression results for equation (3). These estimates focus on the more accurate wall-to-wall distances and include only the sample of California homes for which we can precisely identify neighbor distances using this metric. This sample focuses on homes in denser areas (at least 10 neighbors within a 200-meter radius; see table A3) in fires since 2013 (for older incidents, it is more likely that parcel boundaries have changed since the fire). Figure 5*A* shows effects of the presence of any precode neighbors at various wall-to-wall distances. One or more precode neighbors within 10 meters increases own structure loss probability during a wildfire by about 2.4 percentage points.[25]

[25] Figure A3 summarizes the distribution of neighbors by distance. On average, homes in the sample have 0.43 neighbors with walls between 0 and 10 meters away, and 82% of those neighboring homes are built in the pre–building codes era.

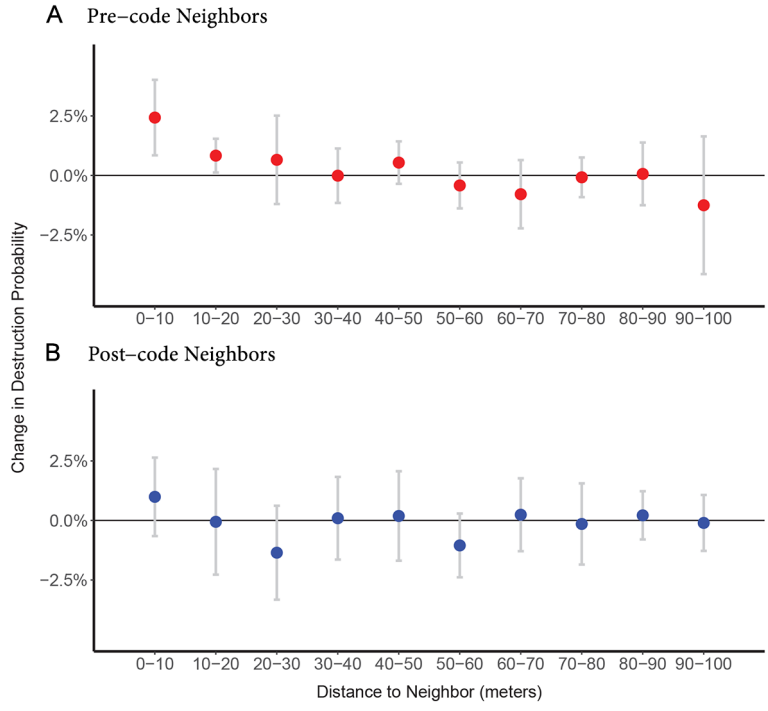


FIG. 5.—Building code neighbor effects by distance. The figure shows coefficients and 95% confidence intervals from a single regression of an indicator for Destroyed on the presence of pre- and postcode neighbors at various distances. *A* shows estimates for indicator variables for the presence of one or more neighbors built without wildfire building codes. *B* shows estimates for indicator variables for the presence of one or more neighbors built after wildfire building codes. The regression also includes own year built (in 4-year bins), street × incident fixed effects, and topographic controls. Distance to neighboring home is wall-to-wall distance.

One or more precode neighbors within 10–20 meters increases own structure loss probability by about 0.8 percentage points, though this estimate is only narrowly statistically distinguishable from zero. Neighbors within 20–30 meters increase risk by 0.7 percentage points, but the confidence interval on this estimate is large and includes zero.

The remaining precode home effects are statistical zeroes beyond 30 meters. Notably, this is similar to the distance of 30 meters that wildfire managers consider to be the home ignition zone: the distance within which flammable material presents a risk of structure ignition (Cohen 2000, 2010; Calkin et al. 2014).[26] The near-zero estimates beyond 30 meters

[26] The larger spillover effects for the very nearest neighbors within the home ignition zone are also consistent with the scientific literature, which identifies a few meters around a structure as particularly critical for structure survival.

bolster the validity of the research design. If our estimates for the nearest neighbors were biased by omitted predictors of resilience that covary within neighborhoods, one would expect that bias to also appear in estimates for homes another few dozen meters away (fig. 1*B* provides a visual illustration of these distances).

Figure 5*B* shows the estimates for postcode neighbors. The confidence intervals for these estimates are wider since we observe fewer postcode homes. However, the point estimates suggest that the presence of close neighbors built under WUI building codes does not substantially increase own structure loss probability. Focusing solely on the magnitudes of the point estimates, we find that the presence of a postcode neighbor within 0–10 meters increases the risk of own home loss by 1.0 percentage points (less than half the 2.4 percentage points we estimate for precode neighbors), but zero is well within the 95% confidence interval.

There is also no implied effect of farther away postcode neighbors on own survival, offering additional placebo evidence to support the identifying assumptions behind this regression. Figure B2 documents sensitivity checks on these estimates. We show there the pattern of estimates we describe for pre- and postcode neighbors is very similar when using the less precise centroid-to-centroid distances on the full set of California homes. We also show that spillover risk from precode neighbors is slightly higher for homes implicated in more destructive incidents (defined as incidents where more homes were at risk). Spillover risk in these incidents remains close to zero for postcode neighbors, suggesting that increasingly destructive fires could magnify the importance of building codes in reducing neighbor-to-neighbor ignition.[27]

Table 2 reports regression estimates for each additional neighbor within 10 meters of the focal home. These estimates provide a key input for the exercise in section VI, where we consider the net benefits of replacing precode homes with postcode homes, including the spillover benefits those replacements confer on their neighbors.[28] Column 1 considers nearby pre- and postcode neighbors as determined by wall-to-wall distances. Each additional precode neighbor increases own structure

[27] Another likely determinant of whether a neighboring home ignites is the direction of local winds that could transport burning material between homes. We do not include wind direction in these specifications for two reasons. First, winds during major conflagrations are highly localized, variable, and difficult to predict or measure. Second, and more importantly, the unpredictability of local winds during wildfires means that there is little that homeowners or policymakers can do to account for these factors in building codes or ex ante mitigation investment decisions.

[28] For the sake of parsimony, this exercise focuses on the effects of neighbors within 10 meters since they represent both the largest and the most precisely estimated spillover effects in fig. 5.

TABLE 2
BUILDING CODE NEIGHBOR EFFECTS ON OWN DESTRUCTION

	(1)	(2)	(3)	(4)
Precode neighbors	.017**	.018***		
	(.007)	(.006)		
Postcode neighbors	.006	.002		
	(.005)	(.007)		
No. of precode neighbors:				
1			.019***	.018***
			(.007)	(.005)
2+			.033**	.037***
			(.015)	(.013)
No. of postcode neighbors:				
1			.012	.006
			(.008)	(.011)
2+			.005	.0004
			(.011)	(.013)
Topography	Yes	Yes	Yes	Yes
Year built bins fixed effects	Yes	Yes	Yes	Yes
Fuel model indicators fixed effects	Yes	Yes	Yes	Yes
Substreet fixed effects	Yes	Yes	Yes	Yes
Distances	Walls	Centroids	Walls	Centroids
Observations	20,757	40,779	20,757	40,779
R^2	.72	.66	.72	.66
Dependent variable mean	.47	.39	.47	.39

NOTE.—The table shows estimates of the effect of having pre- or postcode nearby neighbors on own home destruction. The outcome variable is Destroyed, an indicator for home destruction. The coefficients in cols. 1 and 2 represent the effect of having one additional pre- or postcode nearby neighbor, while the coefficients in cols. 3 and 4 are the effect of having either one or two or more nearby neighbors. "Distances" indicates the method of distance calculation and the subset of homes used. "Walls" distances calculate a nearby neighbor as any home with a wall-to-wall distance of 10 meters or less and are limited to the subset of homes for which we have accurate home footprints. "Centroids" distances calculate nearby neighbors as any homes with adjusted centroid-to-centroid distances of 10 meters or less (see n. 19 for description of adjustment) and use the larger set of homes for which we have parcel boundaries and centroids. Both samples include only homes in code-required jurisdictions in California. Standard errors are clustered by incident.

** $p < .05$.

*** $p < .01$.

loss risk by 1.7 percentage points. The estimated effect of each additional postcode neighbor is 0.6 percentage points and not statistically different from zero. Column 2 shows the same regression using the centroid-to-centroid distances, which are less precise but allow us to include all California homes. The estimated risk posed by additional precode neighbors is slightly larger in this specification, and the estimates for postcode neighbors are smaller at 0.2 percentage points and not statistically different from zero.

Columns 3 and 4 of table 2 repeat the same exercise but replace the counts of pre- and postcode neighbors with indicators for having one or two (or more) nearby neighbors. These estimates show that the effect of one additional precode neighbor roughly doubles the likelihood of

own home destruction in columns 3 and 4, going from 1.9 percentage points to 3.3 percentage points for the sample of homes for which we have wall-to-wall distances and from 1.8 percentage points to 3.7 percentage points for the sample of homes for which we have centroid-to-centroid distances. We do not see the same doubling of effect size for postcode neighbors, which is consistent with postcode homes having limited spillover effects on their neighbors.

C. Broader Housing Market Effects of Wildfire Building Codes

In this section, we present evidence on the housing market impacts of wildfire building codes to complement the results on structure resilience in sections V.A and V.B. In principle, building codes could affect housing demand in regulated areas positively if homeowners are aware of and attentive to the resilience benefits that we document. They could also reduce demand if the codes prohibit aesthetic features that homebuyers value highly (such as cedar siding). These demand effects could also manifest at the neighborhood level through the kinds of housing externalities demonstrated by Rossi-Hansberg, Sarte, and Owens (2010), Fu and Gregory (2019), and others. On the other side of the market, building codes could in principle reduce the supply of new housing if additional required investments substantially increase overall construction costs.[29]

To understand the degree to which local housing markets were impacted by the presence of wildfire building codes, in appendix C we directly examine whether home sale prices and housing growth rates responded to the implementation of building code mandates. To do so, we test for changes in homes sales and growth rates among the larger set of California homes that face some wildfire hazard. Tables C2–C4 show that jurisdictions where wildfire building codes were implemented did not experience meaningful changes in transaction prices or home growth rates after the codes came into effect. It does not appear that California's wildfire building codes led to meaningful shifts in the local housing market. We speculate that the relatively subtle aesthetic changes required by the mandate and the limited understanding by buyers of its resilience benefits help explain the lack of housing market response.

[29] Ostriker and Russo (2022) document such an effect for home elevation requirements in floodplain areas, finding that one-quarter of the reduction of flooding damages comes from housing relocation away from the requirements, while the rest is due to directly reduced damages. In our setting, the cost of mitigation is considerably lower: as we discuss in sec. VI.C, estimates of the total additional cost of to-code construction are generally low relative to the average value of homes.

VI. Net Social Benefits of Adaptation Policies

This section turns to the topics of cost-effectiveness and welfare. We explore two questions: (1) For which homes are the code-required investments cost-effective? (2) Given frictions in take-up, what are the welfare effects of building standards and other possible policies to encourage adoption? Section VI.A develops a model of hazard mitigation investment. Section VI.B derives the welfare properties of policies to encourage investment in this model, focusing on building code mandates and subsidies. Section VI.C discusses how we parameterize the model using our empirical estimates and values from the literature. Section VI.D presents results showing where mitigation investment is cost-effective, and section VI.E compares the welfare effects of various policies to encourage mitigation.

A. *Model of Natural Hazard Mitigation Investment*

Homeowners indexed by i face a risk p_i of a natural disaster in their neighborhood.[30] Conditional on a disaster occurring, they incur loss L with probability q. A homeowner's overall probability of loss is then $p_i q$. They can invest in mitigation (e.g., disaster-resistant construction) at cost m, which reduces their overall probability of loss to $p_i(q - \tau)$. For ease of exposition, this section presents a static model. Our actual calculations discount future benefits over the life of the home at 5% per year.

We say that mitigation by homeowner i is privately cost-effective if the reduction in expected loss outweighs the mitigation cost:

$$p_i qL - [p_i(q - \tau)L - m] > 0 \Rightarrow$$
$$p_i \tau L > m.$$

This measure is likely a lower bound on the true private cost-effectiveness of mitigation given anecdotal evidence that many households are not fully insured against disaster losses (e.g., Klein 2018). Appendix section D.5 describes an expected utility calculation that accounts for additional benefits from reduced exposure to uninsured risk. We focus on the lower bound here because it is more conservative and requires fewer assumptions about the utility function, the completeness of insurance, initial wealth, and other parameters.

[30] Here we build a model that is empirically tractable while capturing the key economic trade-offs of a hazard mitigation mandate. Existing theoretical work—including the model of strategic mitigation of a public bad described in Costello, Quérou, and Tomini (2017)—considers a richer framework. These more general treatments include strategic interactions between agents who make repeated defensive investments against a mobile public bad. We see these strategic interactions as less central in our application, where homeowners are making large one-time investments. Our model also differs by incorporating risk misperception. Given the large empirical literature on misjudged disaster risk, we see this margin as likely important for determining real-world outcomes.

In this benchmark version of the model, homeowner decisions to mitigate will align with a social planner whose goal is to maximize social net benefits. Sections VI.A.1 and VI.A.2 add elements that drive a wedge between some homeowners' choices and what the social planner would choose.

1. Adding Spillovers

Now we assume that mitigation by i creates spillover benefits η for close neighbors. We adopt an empirically tractable specification where investment by i reduces neighbor j's loss probability during a disaster so that the decrease in expected loss for each affected neighbor j is $p_j \eta L$. Letting q_{js} represent the neighbor's baseline loss probability in a disaster, the spillover benefit is $p_j q_j L - p_j (q_j - \eta) L = p_j \eta L$.[31] A social planner would then choose that i mitigate if

$$p_i \tau L + \sum_j p_j \eta L > m.$$

In our empirical application, affected neighbors are close in space (tens of meters). Since there is little meaningful variation in p over such short distances, we simplify the calculation and notation by replacing p_j with p_i and letting δ_i be the number of close neighbors for home i. We can then write that mitigation is cost-effective for society if the following conditions hold:

$$\begin{aligned} & p_i \tau L + \delta p_i \eta L > m \Rightarrow \\ & p_i L(\tau + \delta_i \eta) > m. \end{aligned} \tag{4}$$

We define the break-even level of fire risk p_i where the social planner is indifferent between a homeowner investing and not as $\tilde{p} \equiv m/(L(\tau + \delta_i \eta))$.

2. Adding Risk Misperception

A large literature has consistently shown that households substantially underestimate disaster risks. Surveys that compare homeowners' beliefs

[31] Richer models could allow for more heterogeneity in spillover benefits. An example is the model by Fu and Gregory (2019), who find that amenity spillovers from post–Hurricane Katrina resettlement were nonlinear in the average resettlement rate within a 1-mile radius. In comparison, our model restricts to a constant neighbor externality since (1) few homes have more than one or two neighbors in the relevant distance, limiting the empirical scope for nonlinearity according to take-up rate; (2) the evidence in table 2 is broadly consistent with linear spillovers in that it does not reject equal benefits for the first and second neighbor that mitigate; and (3) the limited historical sample of wildfire-exposed homes means that estimation of more complex spillover functions is limited by statistical power.

about wildfire risk to expert assessments of the same properties consistently find substantial downward bias in beliefs (Brenkert-Smith et al. 2019a, 2019b; Meldrum et al. 2019). Studies using property transaction data show that home prices fall after nearby fires or wildfire information campaigns, consistent with biased beliefs about wildfire risk beforehand (Loomis 2004; Donovan, Champ, and Butry 2007; McCoy and Walsh 2018). Similar evidence for flood and other disaster risks points in the same direction (Gallagher 2014; Bakkensen and Barrage 2021; Wagner 2022). Suppose that the homeowner underestimates their true risk of disaster by a factor θ such that $0 \leq \theta \leq 1$.[32] Now, i invests if

$$\theta p_i \tau L > m. \tag{5}$$

There are two differences between the homeowner's (eq. [5]) and the planner's (eq. [4]) choices to mitigate: the homeowner does not take into account spillover benefits of mitigation or correctly anticipate the size of their expected loss. This means that the threshold value of p_i that determines investment is higher than the social planner would choose. We define the level of fire risk where homeowners are indifferent between investing and not as $\hat{p} \equiv m/\theta\tau L$, which is larger than $\tilde{p}$. Put another way, homeowners who underestimate their own risk and/or do not internalize spillovers may choose not to mitigate when such investment would be socially efficient.

B. Policies to Encourage Mitigation

Given risk misperception and/or spillovers, policies that encourage mitigation may yield social benefits. The theoretically ideal bundle of policies for addressing a combination of externalities and internalities would be a perfectly effective information intervention combined with a Pigouvian tax to address externalities (see, e.g., Allcott, Mullainathan, and Taubinsky 2014). For both political and practical reasons, such a combination of policies is unlikely to be feasible for addressing disaster risk. We consider two second-best policies that are frequently implemented in practice: building codes and subsidies.

1. Building Codes

Total welfare under a building code that mandates mitigation is

$$\sum_i [p_i(\tau + \delta_i \eta) L - m].$$

[32] Homeowners could also underestimate the effectiveness of mitigation τ or the extent of their losses L, or they could be inattentive to their true risk (i.e., the risk would not be salient). In this model, the effect of these would be equivalent.

Recognizing that homeowners facing $p_i > \hat{p}$ mitigate regardless, the change in welfare due to the policy is the sum of the net benefits of adoption by homeowners, with $0 \leq p_i \leq \hat{p}$. It is instructive to write this in three parts:

$$\underbrace{\sum_{i|p_i \in [\tilde{p}, \hat{p}]} [p_i \tau L - m]}_{\text{A}} + \underbrace{\sum_{i|p_i \in [0, \hat{p}]} p_i \delta_i \eta L}_{\text{B}} - \underbrace{\sum_{i|p_i \in [0, \tilde{p}]} [m - p_i \tau L]}_{\text{C}}.$$

Part A is private net benefits to homeowners who benefit from mitigating but fail to invest because of risk misperception, part B is external benefits to neighbors, and part C is private net costs to homeowners for whom mitigation is not socially cost-effective. This formulation captures the inherent trade-off of a mandate: regulators must weigh the benefits of correcting internalities and externalities against the costs of forced adoption by homes with negative social net benefits.

Disaster mitigation differs importantly from many markets where standards have been evaluated, including consumer goods, personal finance, and health (see n. 1). Two primary determinants of social net benefit—local disaster hazard and the number of close neighbors—are readily observable. This creates scope for targeted mandates that may limit the amount of forced inefficient adoption. The benefits of such targeting depend on the geographic granularity with which the regulator can define code areas and the variance of risk within those spatial units. In our empirical application of wildfire risk—and for other disasters like hurricanes, floods, and earthquakes—disaster risk is highly spatially correlated.

2. Subsidies

A popular alternative policy is to subsidize mitigation. We consider the second-best optimal subsidy to address internalities and exernalities in Allcott and Taubinsky (2015). This subsidy equals the sum of the average undervaluation and the average externality for the marginal adopter in the efficient outcome.[33] In our setting, this equates to

$$s = E_{i|p_i(\tau+\delta_i\eta)L=m}[(1-\theta)p_i\tau L + \delta_i \eta L].$$

This amount represents the average shortfall in willingness to pay for households just at the margin of efficient adoption. Ignoring for a moment the cost to finance the subsidy payments, total welfare under the subsidy is

[33] In empirical applications, the discrete nature of the value function maximization problem implies that this formula may differ slightly from the optimal subsidy calculated using, e.g., a grid search over possible subsidy levels. Simulations with our data find that this difference is negligible in our setting.

$$\sum_i 1[\theta p_i L + s \geq m][p_i(\tau + \delta_i)L - m].$$

This subsidy corrects behavior on average but does not necessarily yield the first-best outcome for all homes. Differences in δ_i across homes marginal to the policy lead to under- or overadoption for some households. The total fiscal cost of the subsidies provided is

$$s\sum_i 1[\theta p_i L + s \geq m].$$

As with mandates, the observability of disaster risk and number of neighbors creates scope for targeted subsidies. We also consider a differentiated subsidy that pays each efficient adopter the exact amount by which their private perceived willingness to pay falls short of m. This perfectly differentiated subsidy is

$$s_i = \begin{cases} 0 & p_i(\tau + \delta_i\eta)L < m, \\ \max(0, m - \theta p_i L) & p_i(\tau + \delta_i\eta)L \geq m. \end{cases}$$

The perfectly targeted subsidy yields the first-best level of mitigation investment, with total fiscal cost of subsidies equal to $\Sigma_i s_i$.

C. *Empirical Implementation*

We parameterize our model using data on a large sample of California homes in areas of wildfire hazard, empirical results from our study, and values from the literature. We calculate equation (4) for each home, which allows us to identify where mitigation (i.e., wildfire-resistant construction) is cost-effective. We then calculate equation (5) for each home, which allows us to evaluate take-up in the absence of a mandate or subsidy policy. Finally, we compute the empirical analogs of the welfare expressions for building codes and subsidies in sections VI.B.1 and VI.B.2.

This exercise uses a sample of 1.1 million homes in wildfire-prone areas throughout California. Unlike the empirical analysis of building code effects, which uses homes located inside historical wildfire perimeters, the net benefits calculation considers homes throughout California, not just those that faced a wildfire. Specifically, we include all single-family homes in California for which we can obtain key home characteristics information (year built, square footage) and that are located in a census tract where home average wildfire hazard is at least 0.1%.[34] The rest of this

[34] As a result of these restrictions, 50 of 58 California counties are included in the sample. Imperial, Inyo, Kings, San Francisco, Sutter, and Yolo are excluded because they do not have any tracts with home average wildfire hazard greater than 0.1%. Del Norte and Mendocino are excluded because the assessment data do not include information on square footage or year built for those homes.

section summarizes the calibration of model parameters. We reserve more comprehensive descriptions for appendix D.

Wildfire hazard (p_i).—We identify annual wildfire hazard probability p_i for each home in the sample using the same definition as in section III: it is the product of burn probability (the total annual wildfire probability) and flame length exceedance probability 4 (conditional on any fire, the probability that the fire will reach moderate or greater threat status), both obtained from Scott et al. (2020).

Voluntary take-up without building codes.—Detailed data on wildfire preparedness for homes outside building code jurisdictions are not systematically collected or tracked.[35] We conducted a comprehensive review of survey studies of wildfire home hardening (see table A4). Our benchmark assumption about adoption in the absence of building codes is based on professional assessments of 1,474 Colorado homes in Champ et al. (2020). Of those, 40% use building materials that would comply with California codes in at least two of three areas: roof, exterior siding, and deck.

Own mitigation benefits (τ).—Our estimates of the effect of codes on structure survival in section V can be seen as intention-to-treat estimates of the effect of mitigation investment. Given the above assumption about voluntary take-up, the standard Wald estimator gives τ as the ratio of the reduced-form estimates and the difference in take-up rates between jurisdictions where codes are required and not required.[36] Using our estimate from table 1, column 2, we divide by 1 minus the voluntary adoption rate: $\tau = 0.133/0.6 = 0.22$.

Number of neighbors (δ_i).—We measure each home's number of neighbors directly. We count neighbors within 30 meters of centroid-to-centroid distance, which roughly corresponds to 10 meters of wall-to-wall distance (see n. 19) and is less computationally demanding in this larger sample.

Neighbor benefits (η).—Spillover benefit η is calculated similarly to τ but is the effect of having a nearby postcode neighbor instead of a precode neighbor. Our estimate of the effect is therefore $0.017 - 0.006 = 0.011$ (table 2, col. 1). We divide by 1 minus the voluntary adoption rate: $\eta = 0.011/0.6 = 0.018$.

Losses (L_i).—Assumed losses L_i for a home destroyed by wildfire include rebuilding costs, contents of the home, alternative living costs during rebuilding, debris removal, and hazardous waste cleanup. Per-square-foot rebuilding, contents, and alternative living arrangements costs come from

[35] Remote sensing methods using aerial or satellite imagery are not yet capable of accurately determining the fire readiness of hard-to-observe building components, like siding, eaves, and vents. This may change in coming years, as this is a major area of current investment.

[36] See, e.g., Angrist and Pischke (2009, 127–33). This calculation assumes perfect compliance by homes subject to codes and a homogeneous effect of mitigation on structure survival.

the FEMA (2021) Hazus model. Hazus provides loss cost estimates at the census block level in 2018 US dollars per square foot. These costs vary on the basis of local wages and the mix of construction classes (economy, average, custom, and luxury) in a given block. Appendix section D.1 shows that the 5th and 95th percentile of per-square-foot losses are roughly US$200 and US$350, with a mean of US$260. Total cost L_i is the square footage for home i times these per-square-foot costs, plus a garage rebuilding cost adjustment (also provided by Hazus at the census block level) and a provision for hazardous waste disposal/site cleanup, as described in appendix section D.2.

Mitigation costs (m_i).—Our main estimate of m_i comes from Headwaters Economics (Quarles and Pohl 2018), which uses construction estimating tools to calculate the cost of complying with California's Chapter 7A wildfire code. Overall, the study reports zero cost difference between code-compliant and standard designs because one aspect of code-compliant construction—exterior siding—is substantially less expensive than standard designs. These savings offset increased costs in other areas. Our main estimate of m_i ignores the reported savings from siding since the failure to adopt this less expensive but more resilient option in the baseline scenario may indicate aesthetic preferences that offset the lower construction costs. After making this one adjustment to the estimate from Headwaters Economics (Quarles and Pohl 2018), the resulting code compliance cost is US$15,660 (in 2018 US dollars) for a 2,500-square-foot reference home. In scenarios where we hold reconstruction and mitigation costs constant, we set square footage to 2,500 to match this source as closely as possible. In scenarios where we allow reconstruction and mitigation costs to vary across homes, we convert to a per-square-foot mitigation cost of $15{,}660/2{,}500 = 6.22$.

We also report results for alternative cost estimates from the National Association of Home Builders (NAHB), which reports low and high scenarios of US$7,868 and US$29,429 (Home Innovation Research Labs 2020). We deflate the NAHB estimates to 2018 US dollars to get US$7,634 and US$28,553. Finally, our retrofit scenario uses the estimate of US$62,760 from Headwaters Economics to fully replace roofing and exterior walls on an existing home.

D. Where Is Mitigation Cost-Effective?

This section documents the cost-effectiveness of wildfire hardening investments across at-risk areas across California. This section first considers cost-effectiveness of wildfire hardening investment for a home of constant size and rebuilt cost given varied levels of wildfire risk and number of close neighbors. We then incorporate heterogeneity in size and rebuilding costs.

1. Constant Rebuilding and Mitigation Costs

Figure 6 shows census tract averages of annual wildfire hazard and number of near neighbors. The wildfire hazard reaches strikingly high levels: several tracts face event probabilities above 2% per year, implying a significant wildfire exposure every 50 years on average. The scale shows the social benefit of mitigation investment in each census tract, following equation (4). These estimates assume a typical new home based on the 2,500 square footage considered by Headwaters Economics (Quarles and Pohl 2018). The dashed line shows a threshold for positive net benefits of building standards assuming a typical home based on the 2,500-square-foot cost estimates considered by Headwaters Economics (Quarles and Pohl 2018). Homes to the right of this line have lower expected costs with mitigation investments than without. The threshold bends left as the number of neighbors increases because of the spillover benefits of mitigation. For a home with zero near neighbors—that is, a home that does not have other homes near enough for wildfire risk to spill over to its neighbors—the break-even annual wildfire hazard is 0.40%. This falls to 0.37% with one near neighbor and 0.30% with four near neighbors. Appendix

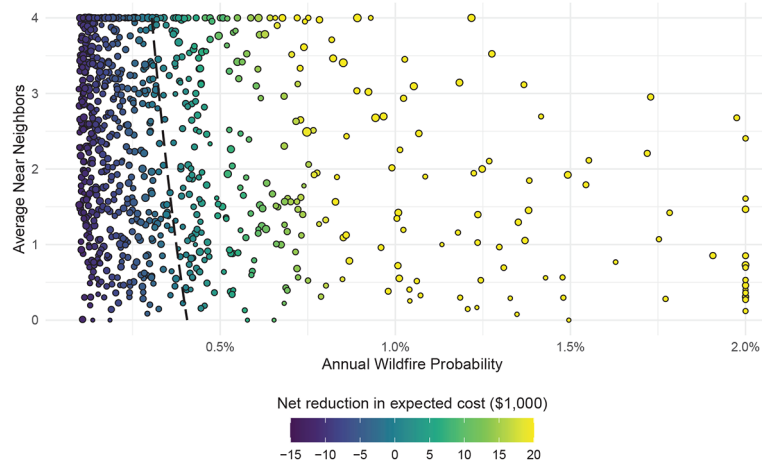


FIG. 6.—Mitigation cost-effectiveness by fire hazard and number of neighbors. The figure plots census tract averages of the cost-effectiveness of mitigation by annual probability of a damaging wildfire and number of close neighbors for California homes in high wildfire hazard areas. Circle shading indicates average net benefits in the census tract using the cost-effectiveness measure, which is a conservative lower bound on total net benefits. Circles indicate zip code averages of annual wildfire hazard from Scott et al. (2020) and represent a snapshot as of 2014. Number of neighbors is the number of homes within a 30-meter centroid-to-centroid distance. Circle size is proportional to the number of homes in the census tract. The dashed line shows a threshold for zero net reduction in expected cost. See text for discussion and alternative scenarios.

TABLE 3
BREAK-EVEN WILDFIRE HAZARD (%) UNDER RISK AVERSION AND ALTERNATIVE COSTS

Cost Estimate (US$)	Source	Insured (%)				
			67		33	
		100	$\gamma = 2$	$\gamma = 5$	$\gamma = 2$	$\gamma = 5$
		A. New Home				
0	HE low	.00	.00	.00	.00	.00
7,634	NAHB low	.16	.16	.14	.13	.08
15,660	HE	.34	.32	.29	.27	.18
28,553	NAHB high	.61	.59	.54	.50	.35
		B. Retrofit				
62,760	HE	1.35	1.30	1.24	1.18	.98

NOTE.—The table shows the estimated minimum annual wildfire probability for which building standards yield positive net benefits under various assumptions about cost, share of losses insured, and risk aversion. Probabilities are reported as percentages (e.g., 0.33% per year). For partial insurance scenarios, γ is the coefficient of relative risk aversion. Calculations assume 2.5 near neighbors. See text for details of these calculations. HE = Headwaters Economics (Quarles and Pohl 2018); NAHB = National Association of Home Builders (Home Innovation Research Labs 2020).

section D.4 documents average cost-effectiveness using the alternative mitigation cost scenarios given in table 3.

These cost-effectiveness estimates are a lower bound on the net benefits of mitigation investments. One reason for this is that many homeowners are not fully insured for wildfire losses. Even for properties covered by homeowners insurance, Klein (2018) reports that coverage limits for wildfire-destroyed properties are often up to 50% below actual monetary losses. Through the lens of a standard expected utility model, the fact that homeowners are not fully insured makes the possibility of a wildfire loss more costly than in the risk-neutral model and thus makes the net benefits of mitigation investment higher than the risk-neutral cost-effectiveness estimates as long as homeowners have any nonzero degree of risk aversion.[37] Table 3 reports break-even annual wildfire probabilities for a home with 2.5 near neighbors (the mean among the all-California sample we use in this section) based on the expected utility model in appendix section D.5, which incorporates measures of risk aversion in the homeowner's utility function. Although this model requires additional strong assumptions, these back-of-the-envelope numbers depict how risk aversion might affect benefits. For example, if mitigation costs US$15,660, a homeowner with a coefficient of relative risk aversion of 5 and an insurance policy covering two-thirds of total losses would be better off investing

[37] We take the positive empirical fact of partial insurance as given. This empirical fact is sufficient to establish that the risk-neutral cost-effectiveness measure is a lower bound on the net benefits of mitigation investments. The question of why homeowners are not fully insured for wildfire risk is a critical question for future work (e.g., Boomhower et al. 2024).

in mitigation wherever the annual probability of a damaging wildfire exceeds 0.29%.[38]

Table 3 also reports results using other estimates of mitigation cost. The zero net cost estimate from Headwaters Economics (Quarles and Pohl 2018) leads to positive benefits for any level of hazard. The two additional estimates from Home Innovation Research Labs (2020) bracket the main cost estimate. Finally, the estimated retrofit cost of US$62,760 results in much higher break-even hazard levels for existing homes. This kind of full retrofit to existing homes appears to generate positive benefits for only a handful of areas with extreme fire hazard. We note that a partial retrofit—that is, a retrofit undertaken during other renovations—would likely imply lower mitigation costs and therefore be cost-effective in more areas.

Beyond risk aversion, wildfire-safe construction likely has additional benefits that are not included in our calculations. These include reductions in public expenditures on firefighting during large wildfires (Baylis and Boomhower 2023), reduced need for public assistance to fire victims (Deryugina 2017), avoided emotional and psychological distress, and less need for public safety power shutoffs that interrupt electricity service during high fire-risk periods.[39] Projected increases in future wildfire probabilities throughout North America suggest that the benefits of mitigation will continue to grow.[40]

2. Heterogeneity in Rebuilding and Mitigation Costs

The cost-effectiveness estimates in section VI.D.1 assume a 2,500-square-foot newly constructed home based on the model home from Headwaters Economics (Quarles and Pohl 2018). This section introduces heterogeneity in home size and per-square-foot rebuilding costs. Table D1 describes the variation in these factors within the homes in our sample of California homes. Per-square-foot rebuilding costs from the FEMA Hazus model vary across census blocks because of both differences in local labor costs and differences in the typical quality of construction (economy, luxury, etc.). The 5th, 50th, and 95th percentiles of per-square-foot rebuilding

[38] Studies of the property insurance market generally report high implied levels of relative risk aversion. Cohen and Einav (2007) and Sydnor (2010) examine deductible choices in auto and homeowners insurance, respectively, and find double-digit values for the mean household across a variety of specifications. Evidence from other markets suggests values closer to the low single digits (e.g., Gertner 1993; Chetty 2006).

[39] For a systematic review of catastrophic wildfire costs, see Feo et al. (2020).

[40] Risk Factor, an open source model of fire risk produced by the nonprofit First Street Foundation, predicts roughly a doubling in the annual probability of homes in wildland areas facing a major wildfire threat in the next 30 years. See https://riskfactor.com/ for details.

costs ("Losses per sf") in our sample are 198, 249, and 354, respectively. As discussed previously, square footage also varies across homes, with more recently constructed homes being larger on average. Table D2 also reports the pairwise correlations between per-square-foot rebuilding costs, square footage, annual wildfire hazard, and number of neighbors.

Figure 7 shows the distribution of structure-level cost-effectiveness using individualized loss value and up-front investment assumptions that depend on both the location and the size of the home. Each of the box and whiskers plots summarizes the distribution of individual cost-effectiveness estimates for homes in a given range of wildfire hazard. Figure 7*A* shows the distribution of net benefits for hardening investments made at the time of construction. For each home, we calculate the net benefits from hardening a newly constructed home with the observed square footage and local rebuilding cost factors. In the highest-hazard areas (annual wildfire hazard above 0.75%), mitigation is cost-effective for the full range of observed losses and mitigation costs. For homes with annual wildfire

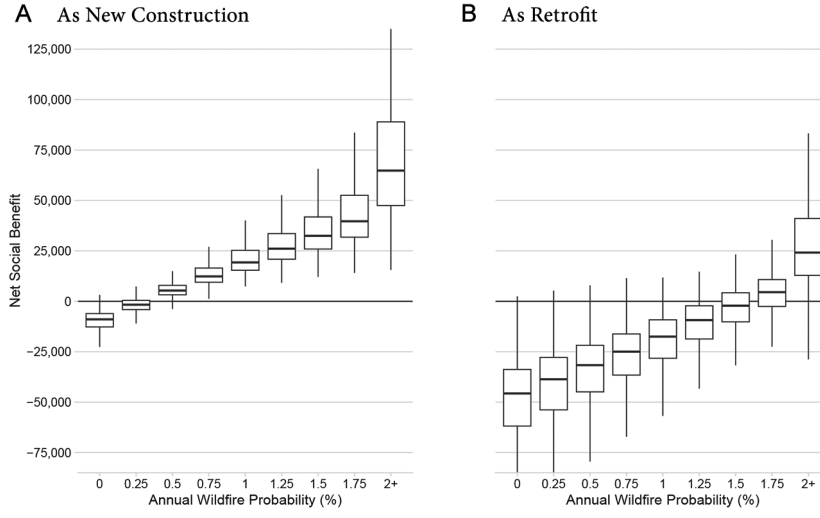


FIG. 7.—Heterogeneity in cost-effectiveness of home hardening. The figure shows box plots of net social benefits (cost-effectiveness) of mitigation for the all-California sample of 1.1 million homes facing some wildfire risk. Net social benefits are calculated using home-specific wildfire hazard, replacement costs, and mitigation costs. Each box indicates homes with similar annual wildfire hazard. Values on the horizontal axis indicate the left side of the interval of wildfire probability; for example, 0.25 indicates homes with wildfire hazard between 0.25% and 0.5%. Whiskers indicate 5th and 95th percentiles, and boxes show the median and interquartile range. *A* uses the central estimate of mitigation costs for newly constructed homes (US$15,660), and *B* uses the estimate of retrofit costs (US$62,760), both from Headwaters Economics (Quarles and Pohl 2018), adjusted for the size of home.

probability between 0.5% and 0.75%, mitigation is cost-effective for over three-quarters of homes. In the lower wildfire hazard bins, relatively few homes pass this risk-neutral cost-effectiveness test. These more detailed heterogeneity-inclusive results are broadly consistent with the constant cost analysis in section VI.D.1, which found that hardening for a typical home was cost-effective in areas with wildfire risk above 0.33%. Figure 7*B* shows the distribution of net benefits for the same group of 1.1 million homes under the assumption that hardening improvements are made as retrofits to already built homes. Because of the much higher cost of retrofitting homes, few homes pass the cost-effectiveness test in this case.

In summary, our empirical estimates and model calculations suggest that investments in wildfire-safe construction yield unambiguous net benefits in the most fire-prone areas of California, especially when homes are clustered closely together such that there are large risk spillovers. For areas with lower fire hazard, the sign of net benefits is more sensitive to modeling choices and the assumed cobenefits of mitigation. Further work on the cost-effectiveness of wildfire mitigation measures in low- and moderate-hazard areas is an important area for research, particularly as overall wildfire probabilities continue to rise.

E. Comparing Policies to Encourage Mitigation

This section considers the welfare effects of policies to increase wildfire-safe construction. We consider two categories of policies that are frequently implemented in practice—building codes and subsidies—as well as a no-policy alternative. In each case, we use the model and data to calculate the relevant welfare expressions from section VI.B. This exercise requires one additional assumption, which is the degree of risk misperception θ. While the literature strongly points to $\theta < 1$, the exact size of the behavioral bias is less clear. Our main approach to calculating θ is to find the degree of misperception that matches the implied voluntary take-up rate for properties in our sample (given our other assumptions) to the 40% take-up rate in Champ et al. (2020). This yields $\theta = 0.5$, which we use throughout the policy comparisons presented here. Appendix section D.7.3 document how alternative assumptions about θ impact the policy comparisons. Our calculations can be easily updated in the future as better data on private demand for home hardening investments become available.

Table 4 reports welfare measures under a range of policies. The metric of central interest is column 1 ("Social Net Benefit"), which is the sum of all homeowners' net benefits under a given policy. Columns 2–6 summarize total private benefits ("Private Net Benefit"), spillover benefits to neighbors ("External Benefit"), the share of all households who adopt ("Total Adoption Rate"), the share of all households who adopt and in

TABLE 4
WELFARE UNDER VARIOUS ADAPTATION POLICIES

Policy	Social Net Benefit (US$ millions) (1)	Private Benefit (US$ millions) (2)	External Benefit (US$ millions) (3)	Total Adoption Rate (4)	Inefficient Adoption Rate (5)	Fiscal Cost (US$ millions) (6)
			A. Benchmarks			
1. No policy	3,872	3,546	326	10.2	.0	0
2. Perfect standard	5,106	4,454	652	29.2	.0	0
3. Perfect subsidy	5,106	4,454	652	29.2	.0	886
			B. Building Codes			
4. Hazard only	5,025	4,393	632	27.9	1.8	0
5. County	2,628	2,044	584	37.9	19.7	0
6. Census block	4,849	4,219	629	28.9	3.5	0
7. Street	4,898	4,267	631	28.7	3.1	0
			C. Subsidies			
8. Uniform	4,704	4,153	550	27.7	4.9	1,508
9. Per square foot	5,097	4,460	638	29.1	.8	2,211

NOTE.—The table shows total social net benefits from investment in wildfire-resistant construction under various policies. The sample includes approximately 1.1 million homes in areas of moderate to very high fire risk throughout California. Each row is a different adaptation policy described in sec. VI, and each column is a welfare metric. Social Net Benefit = total net benefit under given policy; Private Net Benefit = total net benefit of mitigation for each person who mitigates; External Benefit = total benefit to neighbors of homes that mitigate; Total Adoption Rate = percentage of all households that mitigate under policy; Inefficient Adoption Rate = percentage of all households that mitigate in spite of having negative social net benefits under policy; Fiscal Cost = total amount of subsidy paid from regulator to mitigating homeowners.

doing so create negative social net benefits ("Inefficient Adoption Rate"), and the total subsidy payments ("Fiscal Cost").[41]

Panel A of table 4 establishes some benchmarks for comparison. Given our calibrated mitigation costs, benefits, and degree of risk misperception, the total net benefit from home hardening investments in the absence of any policy would be about US$3.9 billion. About 10% of the homes in our sample of wildfire-threatened California homes would adopt. Row 2 shows that a perfect standard where the regulator mandated adoption by all efficient adopters would yield US$5.1 billion in net benefit and a take-up rate of 29%. Row 3 considers an idealized perfectly differentiated subsidy that achieves the first-best outcome by paying each efficient adopter (each home where social net benefit exceeds cost) the exact amount by which their perceived private value lags the cost of adoption (if any). This idealized differentiated subsidy would require payments of about US$900 million.

[41] Appendix sec. D.3 gives more details on the construction of these metrics.

Panel B of table 4 compares outcomes under a range of building code policies, all of which involve zero fiscal cost. Row 4 shows a policy where building code is targeted on only wildfire hazard. This type of policy mimics proposed or adopted building standards for wildfire, flood, and other hazards.[42] We consider a policy that requires adoption wherever the annual probability of wildfire exposure exceeds 0.4%, the welfare-maximizing hazard cutoff in our data for such a policy. Total welfare is about US$100 million less than under the ideal benchmarks. The total adoption rate is about 1.3 percentage points lower, and 1.8% of households are induced to adopt despite negative net benefits. These differences arise because of differences in rebuilding cost across homes with identical levels of wildfire hazard. At the same time, these differences are relatively small. Compared with the no-policy alternative, the hazard-based building code achieves welfare gains of US$1.15 billion, and the idealized standard or subsidy achieves welfare gains of US$1.23 billion.

Rows 5–7 of table 4 impose different geographic restrictions on building codes. We assume that for practical implementation purposes, the building standard must be applied to all or no homes within contiguous spatial units. In each case, we assume that the regulator mandates take-up in spatial units where the total net benefits of universal adoption are nonnegative. A building code applied at the county level would significantly reduce welfare compared with no policy. The total take-up rate is 38%, with slightly over half of those being inefficient adopters where the costs of adoption exceed social benefits. This result occurs because the handful of counties where the standard is imposed include large numbers of inframarginal adopters, who would have mitigated regardless. The primary effect of the standard is to induce take-up among large numbers of households with negative social net benefits.

More geographically tailored mandates do better. Standards at the census block or street level increase welfare compared with the no-policy alternative. Targeting codes by census block or street yields social net benefits 25% larger than under no policy. These policies achieve about 95% of the welfare level under the idealized house-level standard or subsidy. Intuitively, the number of inefficient adopters is higher when codes are applied coarsely (e.g., at the county level) because the variance of disaster risk and number of neighbors is larger in these broad geographies than within smaller units. Table D3 illustrates this point by decomposing variance in wildfire risk within and across counties, census blocks, and streets.

[42] Mapmakers at CAL FIRE define code boundaries based on assessed wildfire risk. See https://osfm.fire.ca.gov/divisions/community-wildfire-preparedness-and-mitigation/wildfire-preparedness/fire-hazard-severity-zones/fire-hazard-severity-zones-map/ for examples. For floods, building requirements often apply to homes in the 100-year flood plain.

The large majority of variation in wildfire risk is between as opposed to within blocks or streets.

Panel C of table 4 considers second-best subsidy policies. A fully uniform subsidy, where the regulator chooses the fixed subsidy amount that maximizes net social benefits, yields about US$300 million less in net benefits than a hazard-only building code and requires around US$1.5 billion in subsidy payments. A per-square-foot subsidy, where the regulator offers a fixed amount that scales with home size, produces better results: total welfare under this second-best subsidy is about US$70 million higher than under the hazard-based building codes. However, the per-square-foot subsidy also implies a fiscal cost of US$2.2 billion in transfers and an attendant efficiency cost determined by the relevant cost of public funds, which we do not directly model here. Moreover, unlike the building code policy, the regulator must know the level of risk misperception θ in order to implement the correct second-best subsidy.

Appendix section D.7 includes additional discussion on four related topics, which we summarize here. First, we compare welfare gains under each policy when spillover benefits are included versus excluded, showing that (1) spillover benefits to neighbors contribute meaningfully to the total welfare improvement but (2) the large private benefits rooted in risk misperception mean that spillovers are not necessary for policy interventions to improve welfare. Second, we show that a more sophisticated regulator can slightly improve the building code standard policy by accounting for inframarginal voluntary take-up when drawing code boundaries. Third, we show how welfare outcomes change as we vary the inattention parameter θ. This sensitivity exercise shows that standards are preferred to no policy for $\theta < 0.95$, that this welfare gap is quite large when θ is small, and that standards and no policy yield similar welfare outcomes for $\theta \geq 0.95$. The welfare difference between standards and a uniform subsidy varies little with respect to θ (ignoring again the fiscal cost of subsidies and attendant cost of financing the transfers). Finally, we consider the implications of potential correlation in risk misperception and wildfire hazard. In general, we find that a building code mandate outperforms a no-policy counterfactual irrespective of correlation between misperception and risk, and the corrective policies are particularly valuable when high-hazard homeowners are more likely to underestimate their true risk.

To summarize, realistically granular wildfire building codes achieve welfare outcomes that beat a no-policy counterfactual and are equal or close to those under highly targeted subsidies, even without accounting for the cost of raising public funds. The reason this occurs is that many of the key margins of heterogeneity that determine the net social benefits of take-up are directly observable by the regulator, especially the occurrence probability of a disaster. Moreover, the spatially correlated nature

of disaster risk means that building standards can be targeted to contiguous areas where the risk faced by most homes exceeds the break-even level.

VII. Conclusion

Efficient investment in adaptation is essential in the face of rapidly accelerating disaster losses. Yet efficient take-up of protective technologies and behaviors is thought to be constrained by spatial externalities, insurance market failures, misperception of risk, and other frictions. The pressing question facing researchers and policymakers is how to best respond to these market barriers. One suite of policies focuses on increasing voluntary take-up through information or subsidies. Another option is to mandate certain investments in hazard areas. These policies may differ substantially in their effects and their political acceptability.

This study contributes evidence on the effects and net economic benefits of a mandatory adaptation policy. We provide the first comprehensive empirical evaluation of California's strict wildfire building codes. The analysis uses a new dataset of property-level data on US homes destroyed by wildfire that was created for this study. The new data combine postfire damage assessment records collected from numerous local and state agencies with property assessment data for all homes, destroyed or surviving, that were within the perimeter of the fire. This combination has three important advantages: it collects and harmonizes previously disparate damage data, it contains a complete record of homes that survive as well as homes that are destroyed, and unlike data for floods and other losses, it is reported at the individual property level. Beyond this study, the new data will enable additional important research on disaster losses.

The empirical analysis in this study is bolstered by our ability to observe differences in building code regimes over time, across jurisdictions within California, and between California and other states. The empirical strategy isolates the effect of building code changes using a fixed effects design that compares outcomes for pre- and postcode homes on the same residential street. This approach narrows the comparison to homes experiencing essentially identical wildfire exposures.

The results show that compared with reliance on voluntary action alone, California's wildfire building codes reduced average structure loss risk during a wildfire by about 15 percentage points, or about a 40% reduction. They also reduced the risk to a close neighbor's home by about 2 percentage points, a 6% reduction. These striking results imply materially different levels of resilience in communities with and without such codes. Moreover, the spatial externalities provide a rationale for public policy intervention even if homeowners were fully informed and rational about wildfire risk.

Having documented these large resilience benefits, we then show how the empirical results can be embedded in an economic model that accounts for mitigation costs, spatial spillovers, and risk preferences. We use our results and other values from the literature to provide a back-of-the-envelope approximation of the minimum annual wildfire risk at which universal mitigation generates positive net benefits. In the most fire-prone areas of California, the calculation shows large net benefits of building codes for new homes. Given the high cost of fully retrofitting existing homes to modern standards, full retrofits do not pass a benefit-cost test in most areas. An important task for future research is to identify individual low-cost investments that can cost-effectively improve the resilience of existing homes in high hazard areas.

Finally, we use the model and data to compare the welfare effects of alternative policy approaches to encourage adaptation. The observability of wildfire risk and number of neighbors along with spatial correlation of disaster risk within neighborhoods means that wildfire building codes targeted with a realistic degree of granularity compare favorably with idealized policy interventions, such as a perfectly differentiated mitigation subsidy.

Other longer-run welfare effects resulting from broader mandates or subsidies outside of the scope of our model are likely salutary as well. Improved overall resiliency should result in wildfires growing more slowly within affected neighborhoods and increase the ability of fire management professionals to effectively target resources to areas of greatest need, both of which should reduce the overall social costs of wildfires. Mandated investment could also help mitigate existing insurance market failures: if wildfire-safe construction can be taken for granted because of regulation, the market may be better able to set prices that match risk (Boomhower 2023). In the context of particularly high levels of risk, managed retreat policies that either restrict future development or incentivize the relocation of current homes may be socially beneficial. We view the assessment of such policies as fruitful areas for future work, although this paper provides suggestive evidence that even areas of substantial wildfire risk can be efficiently occupied if structures are sufficiently resilient.

In summary, the data show that an adaptation mandate substantially improved resilience to wildfires, and a cost-benefit exercise suggests that low take-up without standards is more likely driven by market failures than by fully informed individual decision-making. These results are immediately applicable to policy debates in the United States, Canada, Australia, European Union, and other jurisdictions that are seeking to respond to escalating wildfire risk. More broadly, these findings should be of interest to policymakers and researchers confronting other hazards—like floods, hurricanes, and heat waves—where voluntary take-up of self-protective investments seems to be constrained by similar barriers. As

climate change continues to increase disaster losses, improving our understanding of the role of public policy and market incentives in shaping adaptation is increasingly urgent.

Data Availability

Code replicating the tables and figures in this article can be found in Baylis and Boomhower (2025) in the Harvard Dataverse, https://doi.org/10.7910/DVN/A8MB7B.

References

Alexandre, Patricia M., Susan I. Stewart, Nicholas S. Keuler, Murray K. Clayton, Miranda H. Mockrin, Avi Bar-Massada, Alexandra D. Syphard, and Volker C. Radeloff. 2016. "Factors Related to Building Loss due to Wildfires in the Conterminous United States." *Ecological Applications* 26 (7): 2323–38.

Allcott, Hunt, Sendhil Mullainathan, and Dmitry Taubinsky. 2014. "Energy Policy with Externalities and Internalities." *J. Public Econ.* 112:72–88.

Allcott, Hunt, and Dmitry Taubinsky. 2015. "Evaluating Behaviorally Motivated Policy: Experimental Evidence from the Lightbulb Market." *A.E.R.* 105 (8): 2501–38.

Angrist, Joshua D., and Jörn-Steffen Pischke. 2009. *Mostly Harmless Econometrics: An Empiricist's Companion*. Princeton, NJ: Princeton Univ. Press.

Bakkensen, Laura, and Lint Barrage. 2021. "Going under Water? Flood Risk Belief Heterogeneity and Coastal Home Price Dynamics." *Rev. Financial Studies* 35 (8): 3666–709.

Baylis, Patrick, and Judson Boomhower. 2023. "The Economic Incidence of Wildfire Suppression in the United States." *American Econ. J. Appl. Econ.* 15 (1): 442–73.

———. 2025. "Replication Code for: 'Mandated versus Voluntary Adaptation to Natural Disasters: The Case of US Wildfires.'" Harvard Dataverse, https://doi.org/10.7910/DVN/A8MB7B.

Boomhower, Judson. 2023. "Adapting to Growing Wildfire Property Risk." *Science* 382 (6671): 638–41.

Boomhower, Judson, Meredith Fowlie, Jacob Gellman, and Andrew Plantinga. 2024. "How Are Insurance Markets Responding to Climate Change? Risk Selection and Regulation in the Market for Homeowners Insurance." Working Paper no. 32625, NBER, Cambridge, MA.

Booth, Michael. 2023. "Colorado May Force New Homes in Wildfire-Prone Areas to Adhere to a State Building Code." *Colorado Sun*, February 14. https://coloradosun.com/2023/02/14/colorado-building-codes-wildfires-wildland-urban-interface-bill/.

Bradt, Jacob, and Joseph E. Aldy. 2023. "Private Benefits from Public Investment in Climate Adaptation and Resilience." Working paper.

Brenkert-Smith, Hannah, James R. Meldrum, Pamela Wilson, Patricia A. Champ, Christopher M. Barth, and Angela Boag. 2019a. "Living with Wildfire in La Plata County, Colorado: 2015 Data Report." Res. Note no. RMRS-RN-80, Forest Service, US Dept. Agriculture, Rocky Mountain Res. Station, Fort Collins, CO.

———. 2019b. "Living with Wildfire in Montezuma County, Colorado: 2015 Data Report." Res. Note no. RMRS-RN-81, Forest Service, US Dept. Agriculture, Rocky Mountain Res. Station, Fort Collins, CO.

California Department of Insurance. 2018a. "The Availability and Affordability of Coverage for Wildfire Loss in Residential Property Insurance in the Wildland-Urban Interface and Other High Risk Areas of California." California Dept. Insurance, Los Angeles, CA.

———. 2018b. "File 18-2993: Allstate Indemnity Company." Public Notice no. 2018-24, Insurance Companies Rate Filing, California Dept. Insurance, Los Angeles, CA.

Calkin, David E., Jack D. Cohen, Mark A. Finney, and Matthew P. Thompson. 2014. "How Risk Management Can Prevent Future Wildfire Disasters in the Wildland-Urban Interface." *Proc. Nat. Acad. Sci. USA* 111 (2): 746–51.

Campbell, John Y., Howell E. Jackson, Brigitte C. Madrian, and Peter Tufano. 2011. "Consumer Financial Protection." *J. Econ. Perspectives* 25 (1): 91–114.

Cawley, John, and Christopher J. Ruhm. 2011. "The Economics of Risky Health Behaviors." In *Handbook of Health Economics*, edited by Mark V. Pauly, Thomas G. McGuire, and Pedro P. Barros, 95–199. Amsterdam: Elsevier.

Champ, Patricia A., James R. Meldrum, Hannah Brenkert-Smith, Travis W. Warziniack, Christopher M. Barth, Lilia C. Falk, and Jamie B. Gomez. 2020. "Do Actions Speak Louder than Words? Comparing the Effect of Risk Aversion on Objective and Self-Reported Mitigation Measures." *J. Econ. Behavior and Org.* 169:301–13.

Chetty, Raj. 2006. "A New Method of Estimating Risk Aversion." *A.E.R.* 96 (5): 1821–34.

Cohen, Alma, and Liran Einav. 2007. "Estimating Risk Preferences from Deductible Choice." *A.E.R.* 97 (3): 745–88.

Cohen, Jack D. 2000. "Preventing Disaster: Home Ignitability in the Wildland-Urban Interface." *J. Forestry* 98 (3): 15–21.

———. 2010. "The Wildland-Urban Interface Fire Problem." *Fremontia* 38 (2–3): 16–22.

Cohen, Jack D., and R. Stratton. 2008. "Home Destruction Examination: Grass Valley Fire." Technical Paper no. R5-TP-026b, Forest Service, US Dept. Agriculture, Pacific Southwest Region, Vallejo, CA.

Costello, Christopher, Nicolas Quérou, and Agnes Tomini. 2017. "Private Eradication of Mobile Public Bads." *European Econ. Rev.* 94:23–44.

Dehring, Carolyn A., and Martin Halek. 2013. "Coastal Building Codes and Hurricane Damage." *Land Econ.* 89 (4): 597–613.

Deryugina, Tatyana. 2017. "The Fiscal Cost of Hurricanes: Disaster Aid versus Social Insurance." *American Econ. J. Econ. Policy* 9 (3): 168–98.

Donovan, Geoffrey, Patricia Champ, and David Butry. 2007. "Wildfire Risk and Housing Prices: A Case Study from Colorado Springs." *Land Econ.* 83 (2): 217–33.

FEMA (Federal Emergency Management Agency). 2020. "Building Codes Save: A Nationwide Study." Washington, DC: US Dept. Homeland Security.

———. 2021. "Hazus Inventory Technical Manual." Washington, DC: US Dept. Homeland Security.

Feo, Teresa J., Samuel Evans, Amber J. Mace, Sarah E. Brady, and Brie Lindsey. 2020. *The Costs of Wildfire in California: An Independent Review of Scientific and Technical Information.* Sacramento: California Council Sci. Tech.

Fu, Chao, and Jesse Gregory. 2019. "Estimation of an Equilibrium Model with Externalities: Post-Disaster Neighborhood Rebuilding." *Econometrica* 87 (2): 387–421.

Gallagher, Justin. 2014. "Learning about an Infrequent Event: Evidence from Flood Insurance Take-Up in the United States." *American Econ. J. Appl. Econ.* 6 (3): 206–33.

Gertner, Robert. 1993. "Game Shows and Economic Behavior: Risk-Taking on 'Card Sharks.'" *Q.J.E.* 108 (2): 507–21.
Gibbons, Philip, Linda Van Bommel, A. Malcolm Gill, Geoffrey J. Cary, Don A. Driscoll, Ross A. Bradstock, Emma Knight, Max A. Moritz, Scott L. Stephens, and David B. Lindenmayer. 2012. "Land Management Practices Associated with House Loss in Wildfires." *PLOS One* 7 (1): e29212.
Grad, Shelby. 1996. "O.C. Stopped Warning of 'High-Hazard' for Fire Area." *Los Angeles Times*, October.
Hallstrom, Daniel G., and V. Kerry Smith. 2005. "Market Responses to Hurricanes." *J. Environmental Econ. and Management* 50 (3): 541–61.
Home Innovation Research Labs. 2020. "Cost Impact of Building a House in Compliance with IWUIC." Report no. CR1328-2 12302020, Home Innovation Res. Labs, Upper Marlboro, MD.
Hummel, Michelle A., Robert Griffin, Katie Arkema, and Anne D. Guerry. 2021. "Economic Evaluation of Sea-Level Rise Adaptation Strongly Influenced by Hydrodynamic Feedbacks." *Proc. Nat. Acad. Sci. USA* 118 (29): e2025961118.
Imbens, Guido W., and Donald B. Rubin. 2015. *Causal Inference in Statistics, Social, and Biomedical Sciences*. Cambridge: Cambridge Univ. Press.
Insurance Institute for Business and Home Safety. 2019. "Wildfire Codes and Standards: State-by-State Reference Guide." Tampa, FL: Insurance Inst. Bus. and Home Safety.
Intini, Paolo, Enrico Ronchi, Steven Gwynne, and Noureddine Bénichou. 2020. "Guidance on Design and Construction of the Built Environment against Wildland Urban Interface Fire Hazard: A Review." *Fire Tech.* 56:1853–83.
Jacobsen, Grant, and Matthew Kotchen. 2013. "Are Building Codes Effective at Saving Energy? Evidence from Residential Billing Data in Florida." *Rev. Econ. and Statis.* 95 (1): 34–49.
Klein, Kenneth S. 2018. "Minding the Protection Gap: Resolving Unintended, Pervasive, Profound Homeowner Underinsurance." *Connecticut Insurance Law J.* 25:34–111.
Kousky, Carolyn, Erzo F. P. Luttmer, and Richard J. Zeckhauser. 2006. "Private Investment and Government Protection." *J. Risk and Uncertainty* 33 (1): 73–100.
Kramer, H. Anu, Miranda H. Mockrin, Patricia M. Alexandre, Susan I. Stewart, and Volker C. Radeloff. 2018. "Where Wildfires Destroy Buildings in the US Relative to the Wildland-Urban Interface and National Fire Outreach Programs." *Internat. J. Wildland Fire* 27 (5): 329–41.
Kunreuther, Howard C., and Erwann O. Michel-Kerjan. 2011. *At War with the Weather: Managing Large-Scale Risks in a New Era of Catastrophes*. Cambridge, MA: MIT Press.
Larimer County. 2020. "Residential Requirements: A Guide for the General Contractor or Home Builder." Fort Collins, CO: Larimer County Community Development Division.
Levinson, Arik. 2016. "How Much Energy Do Building Energy Codes Save? Evidence from California Houses." *A.E.R.* 106 (10): 2867–94.
Loomis, John. 2004. "Do Nearby Forest Fires Cause a Reduction in Residential Property Values?" *J. Forest Econ.* 10 (3): 149–57.
Lueck, Dean, and Jonathan Yoder. 2016. "Clearing the Smoke from Wildfire Policy: An Economic Perspective." PERC Policy Series no. 56, Property and Environment Res. Center, Bozeman, MT.
Maclay, C.K. 1997. "State Fire Prevention Law Fizzles." *Contra Costa Times*, August.
McCoy, Shawn J., and Randall P. Walsh. 2018. "Wildfire Risk, Salience and Housing Demand." *J. Environmental Econ. and Management* 91:203–28.

Meldrum, James R., Hannah Brenkert-Smith, Pamela Wilson, Patricia A. Champ, Christopher M. Barth, and Angela Boag. 2019. "Living with Wildfire in Archuleta County, Colorado: 2015 Data Report." Res. Note no. RMRS-RN-79, Forest Service, US Dept. Agriculture, Rocky Mountain Res. Station, Fort Collins, CO.

Miller, Rebecca K., Christopher B. Field, and Katharine J. Mach. 2020. "Factors Influencing Adoption and Rejection of Fire Hazard Severity Zone Maps in California." *Internat. J. Disaster Risk Reduction* 50:101686.

National Association of Home Builders. 2007. "Study of Life Expectancy of Home Components." Washington, DC: Nat. Assoc. Home Builders.

Ostriker, Abigail, and Anna Russo. 2022. "The Effects of Floodplain Regulation on Housing Markets." Working paper.

Paci, James, Matthew Newman, and Tim Gage. 2023. "The Economic, Fiscal, and Environmental Costs of Wildfires in California." Technical report, Gordon and Betty Moore Found., Palo Alto, CA.

Plantinga, Andrew J., Randall Walsh, and Matthew Wibbenmeyer. 2022. "Priorities and Effectiveness in Wildfire Management: Evidence from Fire Spread in the Western United States." *J. Assoc. Environmental and Resource Econ.* 9 (4): 603–39.

Profita, Cassandra. 2021. "The Labor Day Fires Burned Towns and Homes. Oregon Has a Plan to Avoid a Repeat." Oregon Public Broadcasting, September 7.

Quarles, Stephen, Pam Leschak, Rich Cowger, Keith Worley, Remington Brown, and Candace Iskowitz. 2013. "Lessons Learned from Waldo Canyon." Tampa, FL: Insurance Inst. Bus. and Home Safety.

Quarles, Stephen L., and Kelly Pohl. 2018. "Building a Wildfire-Resistant Home: Codes and Costs." Bozeman, MT: Headwaters Econ.

Quarles, Stephen L., Yana Valachovic, Gary M. Nakamura, Glenn A. Nader, and Michael De Lasaux. 2010. "Home Survival in Wildfire-Prone Areas: Building Materials and Design Considerations." ANR Publication no. 8393, Univ. California Agriculture and Natural Resources, Oakland.

Rollins, Matthew G. 2009. "LANDFIRE: A Nationally Consistent Vegetation, Wildland Fire, and Fuel Assessment." *Internat. J. Wildland Fire* 18 (3): 235–49.

Rossi-Hansberg, Esteban, Pierre-Daniel Sarte, and Raymond Owens. 2010. "Housing Externalities." *J.P.E.* 118 (3): 485–535.

SBCC (Washington State Building Code Council). 2024. "Emergency Rule Adopted: Rescinding 2021 Wildland-Urban Interface Code." https://snohomishcountywa.gov/6303/Wildland-Urban-Interface-WUI-Code-Update.

Scott, Joe H., Julie W. Gilbertson-Day, Christopher Moran, Gregory K. Dillon, Karen C. Short, and Kevin C. Vogler. 2020. "Wildfire Risk to Communities: Spatial Datasets of Landscape-Wide Wildfire Risk Components for the United States." Washington, DC: Forest Service, US Dept. Agriculture.

Shafran, Aric P. 2008. "Risk Externalities and the Problem of Wildfire Risk." *J. Urban Econ.* 64 (2): 488–95.

Simmons, Kevin M., Jeffrey Czajkowski, and James M. Done. 2018. "Economic Effectiveness of Implementing a Statewide Building Code: The Case of Florida." *Land Econ.* 94 (2): 155–74.

Simon, Matt. 2018. "The Terrifying Science Behind California's Massive Camp Fire." *Wired*, November 19.

Snyder, Tom. 1995. "New Fire Safety, Roof Regulations Considered: City to Discuss Guidelines Oct. 3." *Orange County Register*, September.

Sommer, Lauren. 2020. "Rebuilding after a Wildfire? Most States Don't Require Fire-Resistant Materials." *Nat. Public Radio*, November 25.

St. Denis, Lise A., Karen C. Short, Kathryn McConnell, Maxwell C. Cook, Nathan P. Mietkiewicz, Mollie Buckland, and Jennifer K. Balch. 2023. "All-Hazards

Dataset Mined from the US National Incident Management System 1999–2020." *Scientific Data* 10:112.
Stewart, George. 1995. "North Tustin Spared from Fire Map." *Orange County Register*, December.
Sullivan, Julie Fate. 1995. "City Spares Area from Fire Hazard Designation." *Los Angeles Times*, October 27.
Sydnor, Justin. 2010. "(Over)Insuring Modest Risks." *American Econ. J. Appl. Econ.* 2 (4): 177–99.
Syphard, Alexandra D., Teresa J. Brennan, and Jon E. Keeley. 2014. "The Role of Defensible Space for Residential Structure Protection during Wildfires." *Internat. J. Wildland Fire* 23 (8): 1165–75.
———. 2017. "The Importance of Building Construction Materials Relative to Other Factors Affecting Structure Survival during Wildfire." *Internat. J. Disaster Risk Reduction* 21:140–47.
Syphard, Alexandra D., and Jon E. Keeley. 2019. "Factors Associated with Structure Loss in the 2013–2018 California Wildfires." *Fire* 2 (3): 49.
Syphard, Alexandra D., Jon E. Keeley, Avi Bar Massada, Teresa J. Brennan, and Volker C. Radeloff. 2012. "Housing Arrangement and Location Determine the Likelihood of Housing Loss due to Wildfire." *PLOS One* 7 (3): e33954.
Troy, Austin. 2007. "A Tale of Two Policies: California Programs That Unintentionally Promote Development in Wildland Fire Hazard Zones." In *Living on the Edge: Economic, Institutional and Management Perspectives on Wildfire Hazard in the Urban Interface*, edited by Austin Troy and Roger G. Kennedy, 127–40. Leeds: Emerald.
Viscusi, W. Kip. 1992. *Fatal Tradeoffs: Public and Private Responsibilities for Risk*. Oxford: Oxford Univ. Press.
Wagner, Katherine R. H. 2022. "Adaptation and Adverse Selection in Markets for Natural Disaster Insurance." *American Econ. J. Econ. Policy* 14 (3): 380–421.
Yost, Walt. 1996. "Accord Douses Controversy on Fire Protection." *Sacramento Bee*, October.